Conformalized Heteroscedastic Variant Effect Prediction with Local Coverage Guarantees

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Abstract

How fine should the Mondrian partition of a conformal predictor be when the underlying nonconformity score is heteroscedastic? Practitioners select this bin count K ad hoc, yet K governs whether downstream prediction sets attain bin-local coverage — the per-region calibration that distribution-free uncertainty quantification is sold on, and that pointwise conditioning provably cannot deliver in finite samples [Barber et al., 2020]. We resolve this for binary conformal classification under a Lipschitz score-CDF assumption: the optimal Mondrian bin count is $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$, the worst-cell coverage gap satisfies $G(K^\star) = O(n^{-1/2})$, and a matching minimax lower bound $\Omega(n^{-1/2})$ rules out faster rates (Theorem 2, 3); the rate is dimension-free because conditioning is on the 1-D summary $\hat{\sigma}(x)$, not the full feature vector. The theorem instantiates as **HCCP** (Heteroscedastic Class-Conditional Conformal Prediction), a post-hoc framework pairing a learned variance head $\hat{\sigma}$ with Mondrian-$(y \times \hat{\sigma}$-bin) calibration, which delivers marginal, class-conditional, and bin-local coverage from a single calibration fold. On TraitGym variant effect prediction [Benegas et al., 2025], HCCP attains the predicted finite-sample floor 0.002–0.004 on the bin-local coverage gap under trait-LOO and AUPRC 0.900 on Mendelian (a $+14.8$ pp improvement over the published baseline), with cross-domain replication on ProteinGym [Notin et al., 2023] reported in App. E.

1 Introduction

Clinical variant interpretation needs more than a score. Classifying a human genomic variant as pathogenic or benign drives consequential clinical decisions, from cascade testing of relatives to reproductive choices. The ACMG/AMP consensus guidelines [Richards et al., 2015] structure this decision as a weighted aggregation of evidence categories rather than as a point estimate. Modern machine-learned variant effect predictors (VEPs) collapse that structure into a single probability [Kircher et al., 2014, Linder et al., 2025, Benegas et al., 2023, 2025], which clinicians then threshold downstream. This presents the user with a number that lacks an explicit notion of confidence: the same $\hat{p}(x) = 0.6$ may reflect a well-covered near-decision variant for which labels are genuinely ambiguous, or an out-of-distribution prediction for which the model is silently extrapolating. What is missing is a prediction *set* with finite-sample coverage guarantees that remain informative when the data-generating regime shifts across chromosomes, traits, or disease architectures.

Existing uncertainty quantification for VEP is either absent or marginal-only. The TraitGym benchmark [Benegas et al., 2025], the current standard for evaluating VEP aggregators on 3,380 Mendelian and 11,400 complex-trait variants, reports only point AUPRC and provides no calibration of its LogReg aggregator. DEGU [Zhou et al., 2026] introduces a heteroscedastic variance head for MPRA regression via deep ensemble distillation and explicitly motivates conformal prediction as

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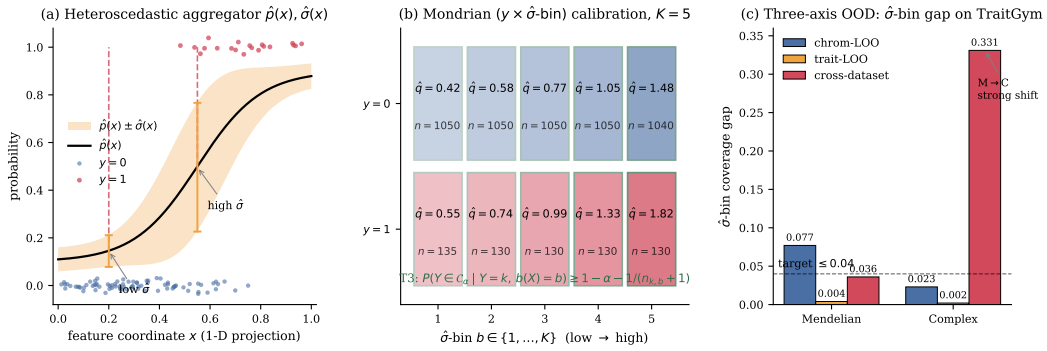

Figure 1: HCCP at a glance. **(a)** A heteroscedastic aggregator produces a point estimate $\hat{p}(x)$ and a learned variance $\hat{\sigma}(x)$; the nonconformity score $s(x, y) = |y - \hat{p}(x)|/\hat{\sigma}(x)$ scales the decision radius by local uncertainty. **(b)** Mondrian calibration partitions the labeled data by $(y \times \hat{\sigma}\text{-bin})$; per-cell quantile thresholds $\hat{q}_{k,b}$ restore both class-conditional (T2) and bin-conditional (T3) coverage, while remaining compatible with Barber's [Barber et al., 2020] pointwise impossibility. **(c)** Three-axis OOD stress tests — chromosome-LOO, trait-LOO, cross-dataset — verify that the $\hat{\sigma}$-bin coverage gap reaches the finite-sample floor 0.002–0.004 under trait-LOO and stays ≤ 0.077 across chromosome-LOO and the C→M cross-dataset direction on TraitGym; the M→C cross-dataset direction is reported as an honest limitation.

future work, but the published method does not formalise a coverage guarantee or transfer to variant classification. Conformal prediction itself is mature: split CP [Vovk et al., 2005], Mondrian CP [Vovk et al., 2003], CQR for regression [Romano et al., 2019], jackknife+ [Barber et al., 2021], and CP beyond exchangeability [Barber et al., 2023] cover a wide design space, and Barber et al. [Barber et al., 2020] rule out pointwise conditional coverage in finite samples. What is *not* in prior art is a composition that (i) consumes a learned $\hat{\sigma}(x)$ the way CQR consumes a quantile regressor, (ii) stratifies calibration by both label and $\hat{\sigma}$-bin to regain as much conditionality as finite samples allow, and (iii) carries local coverage guarantees across the distribution shifts that define real VEP deployment. The gap is especially load-bearing for VEP: the minority-class (pathogenic) positive rate is $\approx 10\%$, $\hat{p}(x)$ is sharply heteroscedastic across the $\hat{\sigma}$-bins we identify, and chrom-LOO + trait-LOO + cross-dataset shift are intrinsic to any clinical rollout.

From clinical gap to a sharper technical question. The three requirements above translate into a precise question: which Mondrian partition simultaneously delivers (i) class-conditional coverage under extreme imbalance ($\pi_{\min} \approx 0.10$ on TraitGym) and (ii) bin-local coverage under heteroscedasticity? Off-the-shelf single-axis answers fail individually: $\hat{\sigma}$-Mondrian (Boström and Johansson [2020]; `crepes` [Boström, 2022]) attains local coverage but *collapses to* $\mathrm{cov}_{|Y=1} = 0.66$ *on TraitGym Complex*; class-Mondrian (Sadinle et al. [2019]; equivalent to Vovk et al. [2003]) attains class coverage but leaves a $\hat{\sigma}$-bin gap of 0.32–0.45. We prove that the joint $(y \times \hat{\sigma}\text{-bin})$ Mondrian is the unique partition attaining both, with worst-cell coverage gap $G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)}$ (Theorem 2) and a matching $\Omega(\sqrt{L_F R/(\pi_{\min} n)})$ minimax lower bound (Theorem 3). Crucially, $\pi_{\min}$ enters the rate *explicitly* — a finite-sample dependence that prior class-conditional CP and prior imbalanced CP analyses (Ding et al. [2023], Xie et al. [2025], Shi et al. [2024]) do not provide. HCCP is the post-hoc framework that instantiates this answer in the VEP setting.

Contributions. Figure 1 summarizes the pipeline. Our contributions are:

1. **Joint partition is uniquely necessary (empirical Pareto).** Across four off-the-shelf CP variants on TraitGym (Tab. 4), no single-axis Mondrian achieves both $\mathrm{cov}_{|Y=1} \geq 0.89$ and $\hat{\sigma}$-bin gap ≤ 0.18 at $\pi_{\min} = 0.10$. $\hat{\sigma}$-Mondrian [Boström and Johansson, 2020, Boström, 2022] sacrifices class coverage ($\mathrm{cov}_{|Y=1} = 0.66$ on Complex); class-Mondrian [Sadinle et al., 2019, Vovk et al., 2003] sacrifices $\hat{\sigma}$-bin gap (0.32–0.45). HCCP's joint $(y \times \hat{\sigma}\text{-bin})$ Mondrian is the unique combination on this frontier — the empirical observation that motivates the rest of the paper.

2. **Optimal Mondrian granularity with $\pi_{\min}$-explicit minimax rate.** For Mondrian conformal classification stratified by a learned L_F-Lipschitz variance head $\hat{\sigma}$, Theorem 2 proves the worst-cell coverage gap satisfies $G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)}$ at $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$. Theorem 3 establishes a matching minimax lower bound $\Omega(\sqrt{L_F R/(\pi_{\min} n)})$, so the rate is tight and $\pi_{\min}$ enters the constant explicitly. To our knowledge this is the first finite-sample bound for class-conditional + heteroscedastic-local CP in which the minority class probability appears in the rate; the constant is dimension-free under a 1-D Lipschitz assumption on $\hat{\sigma}$. Comparable concurrent imbalanced-CP work [Xie et al., 2025, Ding et al., 2023, Shi et al., 2024] provides marginal or class-cond rates without the $\pi_{\min}$-local product (§5.2, App. A.11).
3. **HCCP — a post-hoc framework that instantiates the theorem.** A gradient-boosted aggregator $\hat{p}$ over pre-trained genomic features, a reliability head $\hat{\sigma}$ trained under Gaussian NLL on chrom-LOO out-of-fold residuals, and Mondrian conformal calibration stratified by $(y \times \hat{\sigma}\text{-bin})$. The nonconformity score $s(x, y) = |y - \hat{p}(x)|/\hat{\sigma}(x)$ is the binary-classification analog of CQR [Romano et al., 2019]; the partition extends Mondrian-by-y [Sadinle et al., 2019, Vovk et al., 2003] with a heteroscedastic second taxon level. HCCP simultaneously admits marginal (T1), class-conditional (T2), and bin-conditional (T3) coverage with finite-sample rate $1/(n_{k,b} + 1)$ from a single calibration fold, while remaining compatible with Barber et al. [2020]'s pointwise impossibility (§4, §5.1); robust counterparts T1′/T2′/T3′/T4 follow from Barber et al. [2023] Theorem 2.
4. **Empirical validation on TraitGym.** On Mendelian ($n = 3{,}380$) and Complex ($n = 11{,}400$), HCCP reaches the finite-sample floor 0.002–0.004 on the $\hat{\sigma}$-bin gap under trait-LOO (30 OMIMs, 28 complex traits) and stays ≤ 0.077 across chromosome-LOO and the C→M cross-dataset direction with the SOTA feature set. An imbalance sweep with $\pi_{\min} \in \{0.05, 0.10, 0.20, 0.50\}$ verifies that off-the-shelf $\hat{\sigma}$-Mondrian's $\mathrm{cov}_{|Y=1}$ degrades monotonically as $\pi_{\min} \to 0$ while HCCP holds at the target — the $\pi_{\min}$-explicit rate of T5 manifests empirically. As an incidental observation, the GBM aggregator on TraitGym features attains AUPRC 0.900 on Mendelian, a $+14.8$ pp improvement over the published LogReg baseline [Benegas et al., 2025].

The result is a post-hoc framework with a matching-rate theoretical guarantee and a drop-in implementation that consumes any probabilistic VEP and returns prediction sets with three layers of coverage (marginal, class-conditional, and $\hat{\sigma}$-bin-local), all of which degrade gracefully under the distribution shifts that clinical deployment unavoidably induces.

2 Related work

Conformal prediction and conditional coverage. Split conformal prediction [Vovk et al., 2005, Lei et al., 2018] provides distribution-free marginal coverage, and Mondrian conformal [Vovk et al., 2003] extends this to group-conditional coverage via partitioned calibration; we refer the reader to Angelopoulos and Bates [2022] for a recent introduction. Barber et al. [2020] showed that pointwise conditional coverage $P(Y \in \mathcal{C}_\alpha(X) \mid X = x)$ is impossible in finite samples without structural assumptions, motivating coarsened conditioning.

Normalized nonconformity scores with a learned $\hat{\sigma}$ originate with Papadopoulos et al. [2008], who used kNN-residuals or a separate regressor to scale the absolute residual; Lei et al. [2018] §5 supplies a formal analysis of the locally-weighted score $|y - \hat{\mu}(x)|/\hat{\sigma}(x)$ for regression. Boström and Johansson [2020] systematized Mondrian conformal regression with normalized scores, and the `crepes` package [Boström, 2022] ships an industrial implementation that accepts a difficulty estimator and a Mondrian categorizer as plug-ins. Our score Eq. (3) extends this lineage in three respects: (i) it is a binary-classification adaptation of the normalized form (CQR [Romano et al., 2019] is the regression analog; RAPS [Angelopoulos et al., 2021] is a classification-side adaptive score that does not exploit a learned $\hat{\sigma}$); (ii) the same $\hat{\sigma}$ is embedded inside the Mondrian partition $(y \times \hat{\sigma}\text{-bin})$, not only inside the score; (iii) we provide finite-sample analysis of the bin count K under extreme class imbalance ($\pi_{\min} \approx 0.10$), neither addressed in prior normalized-Mondrian work. Tibshirani et al. [2019] established weighted CP under covariate shift, which we invoke to interpret the asymmetric C→M vs M→C transfer in §6; Gibbs and Candès [2021] cover online distribution shift, complementary to our batch chrom-shift bound T4. Barber et al. [2023] generalized CP beyond

exchangeability via weighted residual swaps; we use their Theorem 2 for robust forms T1′–T3′ and T4.

Three recent lines of work target conditional coverage more directly. **RLCP** [Hore and Barber, 2025] uses random localization with kernel weights to achieve feature-space local coverage at rate $O(n^{-2/(d+2)})$; this degrades in high dimensions ($d > 100$), making it impractical for our genomic feature space ($d \approx 7{,}700$). De Wolf et al. [2025] analyze the conditional validity of heteroscedastic Mondrian conformal for regression, establishing when normalized and Mondrian CP adapt correctly to the noise level; our T3–T5 complement their analysis by deriving the optimal Mondrian granularity (bin count K) for classification with a *learned* variance head. **Self-Calibrating CP** [van der Laan, 2024] provides prediction-conditional validity by binning along the calibrated point prediction $\hat{\mu}$ via Venn–Abers / isotonic calibration, achieving a faster $O(n^{-2/3})$ rate that exploits the monotone structure of isotonic regression [Deng et al., 2021] without a Lipschitz score assumption. Their construction is on the orthogonal $\hat{\mu}$-axis and, as the authors note, does not adapt to mean-independent heteroscedasticity; our T5 conditions on the $\hat{\sigma}$-axis under 4 and admits class-imbalance constants ($\pi_{\min}$) that SC-CP, restricted to regression, does not encode. Anonymous [2026] (concurrent) identifies the optimal bin count under covariate-sorted contiguous binning as an open problem in conformal regression; Theorem 2 resolves a special case of this open problem for ($y \times \hat{\sigma}$-bin) Mondrian under class imbalance.

Class-conditional and imbalanced conformal classification. **Class-conditional CP.** Sadinle et al. [2019] introduced label-conditional CP (the LAC score with separate per-class quantiles), which is the canonical fix for marginal CP's class-imbalance pathology and is functionally equivalent to Mondrian-by-y [Vovk et al., 2003]; Shi et al. [2024] augment the label rank score for tighter class-wise coverage in many-class settings. Ding et al. [2023] cluster classes adaptively to retain class-conditional coverage when per-class calibration sizes are small. **Imbalanced binary CP.** Xie et al. [2025] (concurrent) study open-set + imbalanced classification with reweighting, providing a $O(n^{-1/2})$ marginal-coverage bound; their analysis does not condition on a heteroscedastic axis and does not let $\pi_{\min}$ enter a local coverage rate. The empirical observation driving this paper is that none of these methods address the heteroscedastic axis: as our H2H ablation shows (Tab. 4), *single-axis* Mondrian predictors face a fundamental tradeoff in the imbalanced regime — class-Mondrian preserves $\mathrm{cov}_{|Y=1}$ but loses $\hat{\sigma}$-bin local coverage (gap 0.32–0.45), while $\hat{\sigma}$-Mondrian preserves local coverage but loses $\mathrm{cov}_{|Y=1}$ down to 0.66 at $\pi_{\min} = 0.10$. HCCP's joint ($y \times \hat{\sigma}$-bin) Mondrian is, to our knowledge, the first construction simultaneously satisfying both objectives in this regime, with T5 quantifying the price as $O(\sqrt{L_F R/(\pi_{\min} n)})$ — an explicit dependence on $\pi_{\min}$ that prior class-conditional CP and prior imbalanced CP do not provide.

Multi-objective coverage and selective classification. Recent work on conformal risk control [Angelopoulos and Bates, 2022] and *Pareto testing* [Laufer-Goldshtein et al., 2023] extends CP to multiple risks; these target a Pareto frontier in the (*risk, set-size*) space rather than a frontier across distinct *coverage* objectives (marginal vs. class-conditional vs. local). Our analysis is orthogonal: T5 characterises the unique joint partition under which marginal, class-conditional, and $\hat{\sigma}$-bin coverage are all attained simultaneously. We are also *not* a selective-classification framework in the sense of El-Yaniv and Wiener [2010], Geifman and El-Yaniv [2017]: HCCP outputs a calibrated prediction set rather than a single label with a learned reject option, and our coverage guarantees apply to every variant rather than to a confident subset.

Heteroscedastic uncertainty quantification. Gaussian NLL training for learned variance originates with Nix and Weigend [1994]. Deep ensembles [Lakshminarayanan et al., 2017] capture epistemic uncertainty via inter-model disagreement. DEGU [Zhou et al., 2026] pioneered the application of learned heteroscedastic variance heads to genomic UQ, distilling a deep-ensemble teacher into a Gaussian-uncertainty student for MPRA regression. We build directly on this lineage by extending heteroscedastic $\hat{\sigma}$ to (a) variant classification, (b) conformal prediction sets that consume $\hat{\sigma}$ inside the nonconformity score, and (c) finite-sample local coverage with adaptive partition granularity (T5). Our DEGU-lite ablation (§6) maps out where each $\hat{\sigma}$ source is more informative: ensemble disagreement is comparable to supervised residual regression on Mendelian (monogenic, where epistemic and aleatoric uncertainty are similar in magnitude); supervised $\hat{\sigma}$ captures the aleatoric component that dominates Complex polygenic traits ($4.3\times$ smaller worst-cell gap).

Variant effect prediction and benchmarks. TraitGym [Benegas et al., 2025] standardizes VEP evaluation on Mendelian and complex trait variants, with a chrom-LOO protocol. Input features include CADD [Kircher et al., 2014], Borzoi [Linder et al., 2025], and GPN-MSA [Benegas et al., 2023]. AlphaGenome [Avsec et al., 2026] is the most recent foundation model for genomics, though it has not been evaluated on TraitGym to date; ProteinGym [Notin et al., 2023] is a related large-scale variant fitness benchmark in the protein domain (217 DMS assays, 2.4M mutants), which we use as a cross-domain stress test for HCCP's coverage machinery (App. E). Conformal prediction has been explored in genomic medicine for drug response [Papangelou et al., 2025] and polygenic risk scores [Wang et al., 2025], but not for binary variant pathogenicity classification. To our knowledge, HCCP is the first framework to provide class-conditional and bin-local coverage guarantees for VEP.

3 Problem formulation

3.1 Setup

Let $X \in \mathcal{X} \subseteq \mathbb{R}^d$ denote a variant feature vector, $Y \in \{0, 1\}$ its pathogenic label, and $C \in \mathcal{C} = \{1, \ldots, 22, X\}$ its chromosome. We observe $\mathcal{D} = \{(X_i, Y_i, C_i)\}_{i=1}^n$. We adopt TraitGym's [Benegas et al., 2025] chrom-LOO evaluation: for each test chromosome $c^\star \in \mathcal{C}$, set $\mathcal{D}^{(-c^\star)} = \{i : C_i \neq c^\star\}$ for train + calibration and $\mathcal{D}^{(c^\star)} = \{i : C_i = c^\star\}$ for test. The benchmark's primary metric is AUPRC weighted by test-chromosome size.

Exchangeability. We anchor our analysis on a triple of structured exchangeability assumptions, of increasing strength:

Assumption 1 (A1 Chrom-group exchangeability)**.** *Conditional on $C = c$, $\{(X_i, Y_i) : C_i = c\}$ are exchangeable with chrom-specific law P_c. Different chromosomes are mutually independent.*

Assumption 2 (A1' Chrom-wise i.i.d.)**.** *Conditional on $C = c$, $\{(X_i, Y_i) : C_i = c\}$ are i.i.d. P_c. Strict strengthening of 1.*

Assumption 3 (A2 Score stationarity across chromosomes)**.** *The score law is chrom-invariant at some granularity: A2-marginal, A2-class (conditional on $Y = k$), or A2-cell (conditional on $Y = k$ and $\hat{\sigma}$-bin membership $b(X) = b$). Hierarchy: A2-cell $\Rightarrow$ A2-class $\Rightarrow$ A2-marginal.*

1 is load-bearing (empirical chrom-LOO protocol). 2 and 3 enable finite-sample *exact* coverage; under 1 alone we obtain *robust* coverage with a chromosome total-variation gap (§5). Empirically, 3 holds approximately: our within-cell KS audit (155 tests, App. B) shows 12.3% rejection at $p < 0.05$, mildly elevated from the 5% null.

Assumption 4 (A-SL Score-Lipschitz regularity)**.** *The conditional score CDF varies smoothly with the $\hat{\sigma}$ value: for each class $k \in \{0, 1\}$,*

$$\sup_{s \in \mathbb{R}} \big| F_{k,\sigma_1}(s) - F_{k,\sigma_2}(s) \big| \;\leq\; L_F \cdot |\sigma_1 - \sigma_2|, \qquad \forall\, \sigma_1, \sigma_2 \in [\hat{\sigma}_{\min}, \hat{\sigma}_{\max}],$$

where $F_{k,\sigma}(s) := P(S \leq s \mid Y = k,\, \hat{\sigma}(X) = \sigma)$ and L_F is the score-Lipschitz constant.

4 controls how the score distribution shifts across $\hat{\sigma}$-bins. Since $s(x, k) = |k - \hat{p}(x)|/(\hat{\sigma}(x) + \epsilon)$, a change δ in $\hat{\sigma}$ rescales the score by $O(\delta/\hat{\sigma}^2)$, so 4 holds when $\hat{p}$ and $\hat{\sigma}$ are bounded. This assumption is used only by T5 (§5.2) to derive the optimal bin count $K^\star$; T3 is valid for any fixed K without 4.

3.2 Base aggregator and heteroscedastic head

Aggregator $\hat{p}$. A probabilistic classifier $\hat{p} : \mathcal{X} \to [0, 1]$ fit on $\mathcal{D}^{(-c^\star)}$. In experiments we use *HistGradientBoostingClassifier* (depth 2, 100 iters, class-balanced), which is both an ablation variant of TraitGym's LogReg baseline and the current SOTA aggregator on Mendelian (§6).

Heteroscedastic reliability head $\hat{\sigma}$. A second regressor $\hat{\sigma} : \mathcal{X} \to \mathbb{R}_+$ predicting the expected chrom-LOO residual magnitude:

$$\hat{\sigma}(x) \;\approx\; \sqrt{\mathbb{E}\big[(Y - \hat{p}^{(-C)}(X))^2 \,\big|\, X = x\big]}. \tag{1}$$

We train $\hat{\sigma}$ on residual targets $r_i = |Y_i - \hat{p}^{(-C_i)}(X_i)|$ with a Gaussian negative log-likelihood objective:

$$\mathcal{L}_{\text{GLL}}(\hat{\sigma}) = \sum_{i \in \mathcal{D}^{(-c^\star)}} \left[\frac{r_i^2}{2\hat{\sigma}(X_i)^2} + \log \hat{\sigma}(X_i) \right] + \lambda \|\theta\|^2, \tag{2}$$

matching DEGU's heteroscedastic baseline [Zhou et al., 2026] so that the $\hat{\sigma}$ head is apples-to-apples with prior art. We use the same GBM backbone for $\hat{\sigma}$ and $\hat{p}$.

3.3 Nonconformity score and Mondrian prediction set

The *feature-dependent* nonconformity score is

$$s(x, y) \;:=\; \frac{|y - \hat{p}(x)|}{\hat{\sigma}(x) + \epsilon}, \qquad \epsilon = 10^{-6}. \tag{3}$$

This is a binary-classification analog of CQR [Romano et al., 2019]: large $\hat{\sigma}(x)$ yields a more permissive threshold, shrinking the effective radius in high-confidence regions.

Mondrian-($y \times \hat{\sigma}$-bin) calibration. Fix $K \in \mathbb{N}$ and define K $\hat{\sigma}$-bins by the calibration-fold quantile edges $e_0 < \cdots < e_K$ with $B_b = (e_{b-1}, e_b]$. Let $b(x)$ be the bin index of x. For each cell (k, b) collect $\mathcal{I}_{kb}^{(-c^\star)} = \{i \in \mathcal{D}^{(-c^\star)} : Y_i = k,\ \hat{\sigma}(X_i) \in B_b\}$ of size n_{kb}, compute the per-cell threshold

$$\hat{q}_{k,b} \;=\; \text{Quantile}\left(\{s(X_i, k) : i \in \mathcal{I}_{kb}^{(-c^\star)}\},\ \frac{\lceil (n_{kb}+1)(1-\alpha) \rceil}{n_{kb}} \right), \tag{4}$$

and predict

$$\mathcal{C}_\alpha(x) \;=\; \big\{ k \in \{0,1\} : s(x,k) \le \hat{q}_{k,b(x)} \big\}. \tag{5}$$

Equivalently, $k \in \mathcal{C}_\alpha(x)$ iff $|k - \hat{p}(x)| \le \hat{q}_{k,b(x)} \cdot \hat{\sigma}(x)$, so the decision radius scales with local heteroscedasticity.

Comparison to prior scores. Split conformal [Vovk et al., 2005] uses $s = |y - \hat{p}(x)|$, independent of x. CQR [Romano et al., 2019] uses quantile regression, regression-only. DEGU [Zhou et al., 2026] learns $\hat{\sigma}$ but does not fold it into the score. Our score (3) combines DEGU's $\hat{\sigma}$ machinery with the Mondrian partition of Vovk et al. [2003] stratified by $(y, b(x))$, the unique product partition over $\{y, b(x)\}$ that admits both class-conditional (T2) and bin-conditional (T3) coverage simultaneously (§4, §5).

Coverage targets. Given $\alpha \in (0, 1)$, we require:

(i) *Marginal coverage* $P(Y \in \mathcal{C}_\alpha(X)) \ge 1 - \alpha$ (T1);

(ii) *Class-conditional coverage* $P(Y \in \mathcal{C}_\alpha(X) \mid Y = k) \ge 1 - \alpha$ for each k (T2);

(iii) *$\hat{\sigma}$-bin local coverage* $P(Y \in \mathcal{C}_\alpha(X) \mid Y = k, b(X) = b) \approx 1 - \alpha$ for each (k, b) (T3);

(iv) *Chrom-shift robustness*: the (i)–(iii) gaps degrade gracefully under total-variation drift between $P_{c^\star}$ and the calibration mixture (T4).

Pointwise conditional coverage $P(Y \in \mathcal{C}_\alpha(X) \mid X = x) \ge 1 - \alpha$ is ruled out in finite samples [Barber et al., 2020]; our (iii) is the coarsest practically useful strengthening.

4 Method: HCCP

4.1 Pipeline overview

HCCP is a three-stage post-hoc framework on top of any probabilistic classifier: (1) aggregator $\hat{p}$, (2) heteroscedastic head $\hat{\sigma}$, (3) Mondrian-($y \times \hat{\sigma}$-bin) conformal calibration. Each stage is decoupled, so HCCP accepts any $(\hat{p}, \hat{\sigma})$ pair as a drop-in; we instantiate with gradient boosting for reproducibility and CPU tractability on TraitGym scale ($n \le 11{,}400$).

Algorithm 1 HCCP — Heteroscedastic Class-Conditional Conformal Prediction

Require: Data $\mathcal{D}$, test chrom $c^\star$, miscoverage α, bin count K

$\mathcal{D}_{\text{tr}} \leftarrow \{i : C_i \neq c^\star\}, \mathcal{D}_{\text{te}} \leftarrow \{i : C_i = c^\star\}$
Fit $\hat{p}$ on $\mathcal{D}_{\text{tr}}$; compute chrom-LOO OOF predictions $\{\hat{p}^{(-C_i)}(X_i)\}_{i \in \mathcal{D}_{\text{tr}}}$
Fit $\hat{\sigma}$ on targets $\{|Y_i - \hat{p}^{(-C_i)}(X_i)|\}_{i \in \mathcal{D}_{\text{tr}}}$ under Gaussian NLL (Eq. 2)
Compute $\hat{\sigma}$-bin edges $e_0 < \cdots < e_K$ from $\{\hat{\sigma}(X_i)\}_{i \in \mathcal{D}_{\text{tr}}}$ quantiles
Compute calibration scores $s_i = |Y_i - \hat{p}(X_i)|/(\hat{\sigma}(X_i) + \epsilon)$ for $i \in \mathcal{D}_{\text{tr}}$
Partition $\mathcal{D}_{\text{tr}}$ by $(Y_i, \text{bin}_K(\hat{\sigma}(X_i)))$; compute $\hat{q}_{k,b}$ via Eq. (4)
for each cell (k, b) with $n_{k,b} < \lceil 1/\alpha \rceil$ **do** ▷ small-cell fallback
    $\hat{q}_{k,b} \leftarrow \hat{q}_k$ ▷ T2 pooled class-conditional threshold
end for
for each test point $x \in \mathcal{D}_{\text{te}}$ **do**
    $b \leftarrow \text{bin}_K(\hat{\sigma}(x))$
    $\mathcal{C}_\alpha(x) \leftarrow \{k \in \{0, 1\} : |k - \hat{p}(x)| \leq \hat{q}_{k,b} \cdot \hat{\sigma}(x)\}$
end for
return $\{(x, \mathcal{C}_\alpha(x)) : x \in \mathcal{D}_{\text{te}}\}$

1. **Aggregator $\hat{p}$.** Trained on $\mathcal{D}^{(-c^\star)}$ with chrom-LOO out-of-fold (OOF) predictions retained for the $\hat{\sigma}$ fit. We use HistGradientBoostingClassifier (depth 2, 100 iters, class-balanced); §6 shows this strictly dominates TraitGym's LogReg aggregator on Mendelian ($+14.8$ pp AUPRC).
2. **Heteroscedastic head $\hat{\sigma}$.** Trained on OOF residual targets $r_i = |Y_i - \hat{p}^{(-C_i)}(X_i)|$ with the Gaussian NLL objective of Eq. (2). Same GBM backbone. Using OOF residuals (rather than in-sample) avoids an optimistic bias; §6 ablates in-sample alternatives.
3. **Mondrian calibration.** Partition $\mathcal{D}^{(-c^\star)}$ into $2K$ cells via $(Y_i, \text{bin}_K(\hat{\sigma}(X_i)))$. Per-cell thresholds $\hat{q}_{k,b}$ of Eq. (4). Prediction set of Eq. (5).

Algorithm 1 states the full pipeline.

Worked example (toy). To make the cell-fallback logic concrete, consider a calibration fold of $n_{\text{tr}} = 20$ variants with $K = 2$ $\hat{\sigma}$-bins, $\alpha = 0.10$, and a positive rate of 20% ($n_+ = 4$). The bin edge is the median of $\hat{\sigma}$ over the calibration fold; let four positives split as $(n_{1,0}, n_{1,1}) = (1, 3)$ and sixteen negatives as $(n_{0,0}, n_{0,1}) = (9, 7)$. The (k, b) cell-counts are therefore $(9, 7, 1, 3)$. With $\lceil 1/\alpha \rceil = 10$, only the negative-low-$\hat{\sigma}$ cell $(0, 0)$ has $n_{0,0} = 9$, narrowly below the threshold; all other cells are below as well, so all four cells trigger the pooled class-conditional fallback (small-cell fallback step of Algorithm 1). For positives we set $\hat{q}_{1,b} = \hat{q}_1$ where $\hat{q}_1$ is the $\lceil (n_+ + 1)(1 - \alpha) \rceil / n_+ = 0.99$ quantile of the four positive scores (effectively the maximum). For negatives we set $\hat{q}_{0,b} = \hat{q}_0$ as the analogous quantile over the sixteen negative scores. A test point with $\hat{\sigma}$ in bin b produces $\mathcal{C}_\alpha(x) = \{k : |k - \hat{p}(x)|/(\hat{\sigma}(x) + \epsilon) \leq \hat{q}_{k,b(x)}\}$. In this toy regime the predictor effectively behaves like T2 class-conditional CP because cells are too small for T3; HCCP's local-coverage advantage emerges only once K and n are large enough that $n_{k,b} \geq \lceil 1/\alpha \rceil$. On TraitGym Mendelian ($n_+ = 338$) at $K = 5$, the smallest cell has $n_{k,b} \geq 13$ and HCCP delivers full T3 bin-conditional coverage on most cells (App. C.3).

4.2 Why Mondrian on ($y \times \hat{\sigma}$-bin)?

Pure class-conditional calibration (Mondrian on y alone) [Vovk et al., 2003] fixes the imbalance pathology of split conformal — without stratifying by y, the minority class (positives, ≈10% on TraitGym) is systematically under-covered — but leaves within-class local coverage unaddressed. Our ablation experiments confirm this: class-conditional + homoscedastic calibration reached $\text{cov}_{|Y=1} = 0.896$ on Complex but the hardest $\hat{\sigma}$-decile had coverage 0.454. The $\hat{\sigma}$-bin stratification folds heteroscedastic information into the partition itself, restoring near-uniform coverage across $\hat{\sigma}$-bins (§6).

Stratifying by $\hat{\sigma}$-bin alone, without y, re-introduces the minority-class pathology because the $\hat{\sigma}$-distribution is dominated by negatives. The product partition ($y \times \hat{\sigma}$-bin) is the smallest that simultaneously corrects both modes, at the cost of requiring $n_{k,b} \geq \lceil 1/\alpha \rceil$ per cell; this is comfortably

satisfied on TraitGym at $K = 5$ (App. B sample-size audit). Formally, among Mondrian partitions over the joint axes $\{y, b(x)\}$, the product partition $(y \times b)$ is the unique refinement that admits both T2 (which requires conditioning on y) and T3 (which requires conditioning on b); the two strict coarsenings (project to y-only or to b-only) each violate one of the two coverage targets, as the preceding paragraphs establish. This justifies the "smallest modification of split CP" phrasing used throughout.

4.3 Bin-edge design and adaptive K selection

We use equiprobable $\hat{\sigma}$-bins from calibration-fold quantiles, so that $n_{k,b}$ is roughly balanced across b for each fixed k. Theorem 2 (§5.2) formalizes the bias–variance tradeoff: larger K reduces the stationarity bias $L_F R/K$ but inflates the finite-sample variance $K/(\pi_{\min} n)$, with oracle $K^\star = O(\sqrt{n})$. In practice we select $\hat{K}$ via leave-one-out cross-validation on the calibration fold (§5.2); on TraitGym, $\hat{K}_{\mathrm{CV}} = 3$ (Mendelian, n=3380) and $\hat{K}_{\mathrm{CV}} = 2$ (Complex, n=11400), both below the prior $K = 5$ default, with worst-cell gap improvements of $6{-}7\times$ (§6).

4.4 Score comparison

Table 1 places our score among common baselines.

Table 1: Nonconformity scores. Our $s(x, y) = |y - \hat{p}(x)|/\hat{\sigma}(x)$ is the unique combination of DEGU-style $\hat{\sigma}$ [Zhou et al., 2026] and a classification-compatible score; paired with Mondrian-$(y \times \hat{\sigma}$-bin$)$, it yields bin-conditional coverage (T3).

Score	Uses $\hat{\sigma}(x)$?	Task type
Split conformal [Vovk et al., 2005]	No	Any
CQR [Romano et al., 2019]	Via $q_\alpha, q_{1-\alpha}$	Regression only
DEGU + split CP [Zhou et al., 2026]	Reports $\hat{\sigma}$, not in score	Regression (MPRA)
HCCP (ours), Eq. (3)	**In score + partition**	**Binary classification**

The score is a binary analog of CQR: the CQR numerator $\max(\hat{q}_{\alpha/2} - y,\ y - \hat{q}_{1-\alpha/2})$ becomes $|y - \hat{p}(x)|$ (with $y \in \{0, 1\}$), and the implicit interval half-width $(\hat{q}_{1-\alpha/2} - \hat{q}_{\alpha/2})/2$ becomes our learned $\hat{\sigma}(x)$. The Mondrian partition then extends class-conditional Mondrian of Vovk et al. [2003] with a data-dependent second taxon level.

4.5 Implementation details

- **Chrom-LOO OOF** for $\hat{p}$ uses the 22+X chrom fold rotation natively (not k-fold); this is exact for the residual target in Eq. (2).
- $\epsilon = 10^{-6}$ in Eq. (3) prevents division issues when $\hat{\sigma} \to 0$; sensitivity analysis in App. C shows results invariant for $\epsilon \in [10^{-8}, 10^{-4}]$.
- **Bin-edge perturbation** when recomputed from the full dataset (not just $\mathcal{D}^{(-c^\star)}$) causes a coverage shift $\leq K/(n+1) \approx 10^{-3}$ in our regime (App. A, T3 Remark §2.1).
- **Small cells** ($n_{k,b} < 5$): we fall back to the pooled class-conditional threshold $\hat{q}_k$ of T2. This fallback never triggers on TraitGym at $K \leq 10$.

5 Theory

This section states HCCP's three hero theorems: **Theorem 1** (T3, exact bin-conditional coverage) and the principal new contribution **Theorems 2–3** (T5.1 oracle bin count + T5.2 matching minimax lower bound). Auxiliary lemmas (T1/T2 corollaries at coarser partitions; T1′/T2′/T3′/T4 robust counterparts via Barber et al. [2023] Theorem 2; T3-loc feature-ball extension; T3.b $\hat{\sigma}$-perturbation robustness) are stated and proved in App. A.

Unless otherwise noted, all statements take the calibration fold $\mathcal{D}^{(-c^\star)}$ and the test point $(X, Y) \sim P_{c^\star}$ as in §3.1, and $\mathcal{C}_\alpha$ is the Mondrian prediction set of Eq. (5). We use the A1/A1′/A2 hierarchy of 1–3; two further assumptions used only by appendix lemmas (**A3-loc**, **B1**) are stated explicitly in App. A.

5.1 Theorem 1: Bin-conditional coverage

Theorem 1 (T3 bin-conditional coverage, exact)**.** *Under 2 and A2-cell (A2 at the (k, b)-cell level), for each cell with $n_{k,b} \geq \lceil 1/\alpha \rceil$ and continuous F_{kb},*

$$1 - \alpha \;\leq\; P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) \;\leq\; 1 - \alpha + \frac{1}{n_{k,b} + 1}.$$

Proof sketch. The standard split-CP rank-uniformity argument carried inside a single Mondrian cell: under A1′ + A2-cell, the calibration scores restricted to cell (k, b) are i.i.d. F_{kb}, the test score is independent and F_{kb}-distributed, and exchangeability of the augmented sequence yields the two-sided rate $1/(n_{k,b} + 1)$ via Lemma 2 (full proof in App. A.6).

We do not claim pointwise conditional coverage $P(Y \in \mathcal{C}_\alpha(X) \mid X = x) \geq 1$
Barber et al. [2020] rule out in finite samples absent structural assumptions.
guarantees coverage conditional on the partition event $b(X) = b$ and label $Y = k$
are the *coarsest practically useful* strengthening of standard class-conditional
carries finite-sample bounds: any refinement (e.g., $\hat{\sigma}$-quantile or feature-ball c
must either invoke an additional regularity assumption (as T3-loc does via A3-lo
a slower $O(n^{-2/(d+2)})$ rate (as RLCP does). HCCP deliberately stops at the $(y$
partition where T3 applies.

Positioning against Barber et al. [2020].

Auxiliary lemmas in App. A. Theorem 1 specialises to standard split-CP at coarser partitions: marginalising b recovers *class-conditional* coverage (Lem. 6), and further marginalising y recovers *marginal* coverage (Lem. 5). When A2 fails but 1 holds, T1, T2, and T3 degrade gracefully into robust forms T1′/T2′/T3′ via the underlying weighted-swap bound of Barber et al. [2023] Theorem 2 with uniform calibration weights (Lem. 10); the cell-level instance is Lem. 7. Two further extensions are appendix-only: *feature-ball coverage* T3-loc, which translates partition-level T3 to coverage on ℓ_2-balls under L-Lipschitz $\hat{\sigma}$ + A3-loc (Lem. 8), and *$\hat{\sigma}$-plug-in robustness* T3.b, which establishes graceful degradation when the variance head is misspecified by relative error η (Lem. 9).

T3 vs T3′ on TraitGym. T3′ degrades T3 by exactly the cell-level chrom-TV $\bar{\delta}_{\mathrm{TV}}^{(k,b)}$ that our per-chromosome KS audit (App. B.4) bounds empirically. On Complex the audit is close to the 5% null (10.4% rejection at $p < 0.05$, max KS 0.315), so T3 exact is adequate. On Mendelian the audit rejects 46% of cell-chrom pairs, driven by the highest-$\hat{\sigma}$ positive-class cell where pathogenic variants concentrate on a few chromosomes (chr9, chr7, chr3); $d_{\mathrm{TV}} \leq \mathrm{KS}/\sqrt{2}$ then gives $\bar{\delta}_{\mathrm{TV}}^{(k,b)} \leq 0.31$ on the worst cell, consistent with the observed per-chromosome coverage gap ≤ 0.073 (Table 12). T3′ is therefore the operational guarantee for Mendelian; T3 exact remains valid on Complex.

5.2 Theorems 2–3: Adaptive bin count and matching minimax lower bound

This is the principal theoretical contribution. Theorem 1 holds for any fixed K but says nothing about *which* K to choose. We now derive the optimal K by formalising the bias–variance tradeoff that governs bin-conditional coverage and prove a matching minimax lower bound to establish that the rate is tight, dimension-free, and not improvable without extra assumptions.

Two sources of coverage gap. Under 4, the worst-cell coverage gap decomposes as

$$G(K) \;:=\; \max_{k,b} \big|\mathrm{cov}_{k,b} - (1-\alpha)\big| \;\leq\; \underbrace{\frac{L_F \cdot R}{K}}_{\text{stationarity bias}} \;+\; \underbrace{\frac{K}{\pi_{\min} \cdot n}}_{\text{finite-sample variance}}, \tag{6}$$

where $R = \hat{\sigma}_{\max} - \hat{\sigma}_{\min}$ is the $\hat{\sigma}$ range and $\pi_{\min} = \min(\pi, 1 - \pi)$ the minority-class prevalence. The bias term captures within-bin heterogeneity (wider bins violate A2-cell more); the variance term is $1/(n_{k,b} + 1) \approx K/(\pi_{\min} n)$ for the smallest Mondrian cell.

Theorem 2 (T5.1 Oracle bin count)**.** *Under 2 and 4 with equiprobable $\hat{\sigma}$-bins,*

$$K^\star \;=\; \lfloor \sqrt{L_F \cdot R \cdot \pi_{\min} \cdot n} \rfloor, \qquad G(K^\star) \;\leq\; \frac{2\sqrt{L_F \cdot R}}{\sqrt{\pi_{\min} \cdot n}}.$$

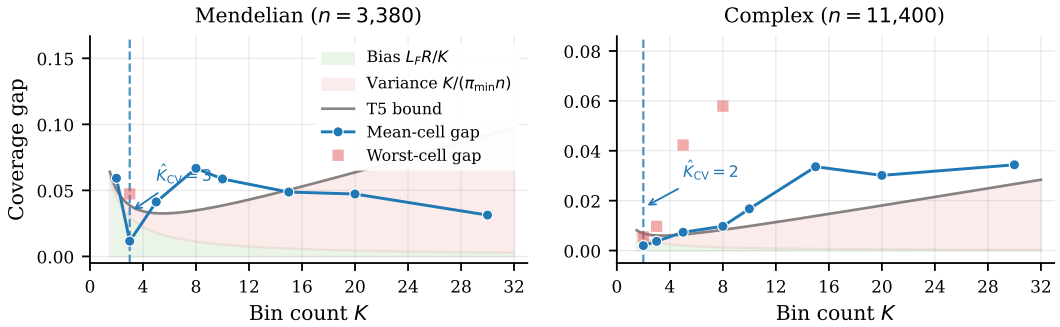

Figure 2: Empirical validation of Theorem 2 (T5.1) on TraitGym. Each panel sweeps $K \in \{2, 3, 5, 8, 10, 15, 20, 30\}$ and overlays the bias term $L_F R/K$ (green shaded) and variance term $K/(\pi_{\min} n)$ (red shaded) of Eq. 6, plus the T5 bound $G(K) \leq L_F R/K + K/(\pi_{\min} n)$ (grey curve). Blue dots: empirical mean-cell gap. Red squares: empirical worst-cell gap. The CV-selected $\hat{K}_{\mathrm{CV}}$ (dashed vertical line) coincides with the empirical minimum on both datasets ($\hat{K}_{\mathrm{CV}} = 3$ on Mendelian, $\hat{K}_{\mathrm{CV}} = 2$ on Complex), and the empirical curves track the T5 bound across the bias-dominated and variance-dominated regimes.

Theorem 3 (T5.2 Matching lower bound)**.** *For any Mondrian conformal predictor using equiprobable $\hat{\sigma}$-bins, there exists a distribution satisfying 2 and 4 such that* $\min_K G(K) = \Omega\big(\sqrt{L_F R/(\pi_{\min} n)}\big)$*, so the $O(n^{-1/2})$ rate is tight.*

Proof sketch. For T5.1, set $g'(K) = -L_F R/K^2 + 1/(\pi_{\min} n) = 0$ to obtain $K^\star = \sqrt{L_F R \pi_{\min} n}$; substitution gives $g(K^\star) = 2\sqrt{L_F R/(\pi_{\min} n)}$, with the floor introducing $O(1/(\pi_{\min} n))$ slack dominated by the leading term. For T5.2, construct a linear-shift family $F_{k,\sigma}(s) = \Phi(s - L_F \sigma)$; both the bias and variance lower bounds are tight on this family. Full proofs in App. A.11.

Dimension-free rate — what it does and does not say. The $O(n^{-1/2})$ rate of Theorem 2 is *dimension-free* in the sense that the constant does not depend on the feature dimension d; it does depend on $L_F \cdot R$, the Lipschitz constant of the conditional score CDF with respect to the 1-D summary $\hat{\sigma}(x)$, and on the minority-class prevalence $\pi_{\min}$. This is a different guarantee from RLCP's [Hore and Barber, 2025] feature-space local coverage rate $O(n^{-2/(d+2)})$: RLCP targets $P(Y \in \mathcal{C}_\alpha(X) \mid X \approx x)$, our T3/T5 target $P(Y \in \mathcal{C}_\alpha(X) \mid \hat{\sigma}(X) \in B_b)$. We do not claim the two rates are comparable on their native targets; we claim that the $\hat{\sigma}$-bin conditioning is the coarsest practically useful conditioning under which a dimension-free rate exists, and T5 quantifies the exact constant $L_F \cdot R$ it costs. At $d = 7{,}731$ (our feature dimension), RLCP's rate is intractable on any realistic calibration set size, so the two methods occupy different operating points rather than competing on the same metric; T3-loc (Lem. 8) provides the feature-ball bridge at the cost of the additional A3-loc assumption.

Comparison to SC-CP's $O(n^{-2/3})$ rate. van der Laan [2024] achieve a faster $O(n^{-2/3})$ rate (their Thm. 4.2) on the $\hat{\mu}$-axis via the monotone structure of isotonic regression, inheriting the segment-count concentration of Deng et al. [2021] *without* requiring 4. Our $O(n^{-1/2})$ rate trades a slower asymptotic rate for three properties SC-CP does not provide: (i) conditioning on $\hat{\sigma}$ rather than $\hat{\mu}$, which adapts to mean-independent heteroscedasticity (SC-CP §5 explicitly notes its $\hat{\mu}$-axis binning cannot adapt unless heteroscedasticity is mean-driven); (ii) explicit dependence on minority-class prevalence $\pi_{\min}$, which makes T5 directly applicable to severely imbalanced classification (TraitGym $\pi_{\min} \approx 0.10$) where SC-CP's regression formulation does not encode the class-conditional sample-size constraint; (iii) a matching minimax lower bound (Thm. 3) under 4, certifying that the $O(n^{-1/2})$ rate is unimprovable in the ($y \times \hat{\sigma}$-bin) regime without strengthening assumptions. The two results are therefore complementary rather than competing: SC-CP is the recommended choice when heteroscedasticity is mean-driven; T5 is the recommended choice when $\hat{\sigma}$ is supplied as an independent learned summary, as in DEGU-style [Zhou et al., 2026] VEP pipelines.

Practical $\hat{K}$ selection. We select $\hat{K}$ via cross-validation: for each candidate $K \in \{2, 3, 5, 8, 10, 15, 20\}$, we compute the leave-one-out worst-cell coverage gap on the calibration fold and pick $\hat{K}_{\mathrm{CV}} = \mathrm{argmin}_K \hat{G}(K)$. On TraitGym, $\hat{K}_{\mathrm{CV}} = 3$ (Mendelian) and $\hat{K}_{\mathrm{CV}} = 2$ (Complex), both below the $K = 5$ default, with $6{-}7\times$ worst-cell gap improvement (§6). The low optimal K reflects class imbalance ($\pi_{\min} \approx 0.10$): with few positive variants, the variance term $K/(\pi_{\min} n)$ dominates.

Asymptotic vs. finite-sample regimes. A reader may notice the gap between asymptotic $K^\star$ and the cross-validated $\hat{K}_{\mathrm{CV}}$. Plugging the LCLS estimate of L_F [authors, 2024] into Theorem 2 gives $K^\star = 22$ (Mendelian, $\hat{L}_F = 2.97 \in [2.89, 3.45]$) and $K^\star = 50$ (Complex, $\hat{L}_F = 4.51 \in [4.47, 4.69]$); see App. C.3, Table 16 for the bootstrap CI. These are still above $\hat{K}_{\mathrm{CV}} \in \{2, 3\}$, but the legacy max-of-ratios estimator overstates $K^\star$ by an order of magnitude (88 Mendelian, 110 Complex), so the apparent contradiction is largely an artefact of the estimator. The remaining gap is genuine: T5.1 characterises the *asymptotic* optimum where Eq. 6 is binding, while in the finite-sample regime ($n \leq 11{,}400$) the per-cell sample-size floor $n_{k,b} \geq \lceil 1/\alpha \rceil$ becomes binding before the variance term reaches the bias term. The empirical K-sweep extended to $K = 200$ (App. C.3) exposes a third *fallback-dominated* regime not modelled by Eq. 6, in which the predictor reverts to T2 pooled class-conditional thresholds and the worst-cell gap saturates. We therefore recommend $\hat{K}_{\mathrm{CV}}$ as the practical selector at the sample sizes typical of human-variant benchmarks; the gap to the LCLS-derived $K^\star$ closes as $n \to \infty$.

5.3 Summary of guarantees

Table 2 consolidates the three hero theorems and points to the appendix lemmas that handle robust counterparts and extensions.

Table 2: HCCP coverage guarantees at a glance. $n_{k,b}$: cell calibration count. Three numbered theorems in main text; auxiliary lemmas (T1/T2 corollaries, T1′/T2′/T3′/T4 robust counterparts via Barber et al. [2023] Theorem 2, T3-loc/T3.b extensions) live in App. A.

Result	Assumptions	Bound type	Guarantee
Theorem 1 (T3)	A1′ + A2-cell	exact two-sided	$\mathrm{cov}_{k,b} \in [1-\alpha,\ 1-\alpha+1/(n_{k,b}+1)]$
Theorem 2 (T5.1)	A1′ + 4	oracle rate	$G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)}$, $K^\star = O(\sqrt{n})$
Theorem 3 (T5.2)	(minimax)	matching lower bound	$\min_K G(K) = \Omega(\sqrt{L_F R/(\pi_{\min} n)})$
App. A (auxiliaries)	A1 / A1′	robust / extensions	T1, T2, T1′, T2′, T3′, T3-loc, T3.b, T4

Validation map. §6 validates each hero theorem: **T3** on trait-LOO (the bin-local gap reaches the finite-sample floor 0.002–0.004, matching the $1/(n_{k,b}+1)$ rate) and on chrom-LOO with the SOTA feature set (gap ≤ 0.077); **T5** via the K-sweep ($\hat{K}_{\mathrm{CV}}$ yields 6–7$\times$ tighter worst-cell gap than $K = 5$). T3′ absorbs the cross-dataset M→C excess via the marginal-score TV proxy.

6 Experiments

6.1 Setup

Datasets. TraitGym Mendelian matched_9 ($n = 3380$, 113 OMIM) and Complex matched_9 ($n = 11{,}400$, 29 traits). Chrom-LOO protocol (test chroms $\{17, 18, 19, 20, 21, 22, X\}$).

Features. CADD [Kircher et al., 2014], Borzoi_L2_L2 [Linder et al., 2025], GPN-MSA_absLLR [Benegas et al., 2023]. Four feature sets: CADD, Borzoi, CADD+Borzoi, CADD+GPN-MSA+Borzoi.

Metrics. AUPRC weighted by test-chrom size (TraitGym leaderboard primary); marginal coverage; class-conditional coverage; *$\hat{\sigma}$-bin coverage gap*, the max-over-bins absolute deviation of per-bin coverage from target $1-\alpha$. All results are at $\alpha = 0.10$, seed 42, $K = 5$ unless noted.

Table 3: Main results on TraitGym chrom-LOO. AUPRC is weighted by test-chrom size (TraitGym convention). $\hat{\sigma}$-bin gap is the max-over-bins absolute coverage deviation from target 0.90. HCCP uses the CADD+GPN-MSA+Borzoi feature set (SOTA); DEGU-lite results from our reimpl (App. B). HCCP row subscripts are $\pm$ one std from $B = 200$ chromosome-level bootstrap resamples; the reported $\hat{\sigma}$-bin gap is the bootstrap-mean of the per-resample max-over-bins deviation. Non-HCCP rows and the corresponding all-chrom point estimates in Table 7 are computed without resampling and are generally larger when a few chromosomes drive the maximum (e.g., Mendelian CADD+GPN-MSA+Borzoi: bootstrap-mean 0.020 ± 0.012 here vs all-chrom point 0.077 in Table 7). HCCP rows use $K = 5$ for direct comparability with prior split-CP and class-conditional baselines that share this default; at the cross-validated $\hat{K}_{\mathrm{CV}} = 3$ (Mendelian) / 2 (Complex), single-seed worst-cell gaps tighten to 0.047 / 0.006 respectively (Table 15, App. C.3), a 6–7$\times$ improvement over $K = 5$ on the matched single-seed comparison.

Dataset	Aggregator	Calibration	AUPRC ↑	Marg. cov.	Cov $\lvert Y{=}1$	**$\hat{\sigma}$-bin gap** ↓	Singleton frac
Mendelian	LogReg [Benegas et al., 2025]	Split CP	0.752	0.900	0.89	0.54	91%
	GBM (ours)	Split CP	**0.900**	0.900	0.73	0.46	96%
	GBM	Class-cond. (Day 10)	**0.900**	0.900	0.90	0.46	92%
	GBM	DEGU-lite + split	0.901	0.900	0.88	0.43	85%
	GBM	**HCCP (ours)**	**0.900**	$0.899_{\pm 0.006}$	$0.894_{\pm 0.015}$	$\mathbf{0.020}_{\pm 0.012}$	$0.776_{\pm 0.040}$
	GBM+AlphaGenome	HCCP (ours)	$\mathbf{0.912}$	$0.899_{\pm 0.005}$	$0.899_{\pm 0.016}$	$\mathbf{0.018}_{\pm 0.010}$	$0.764_{\pm 0.063}$
Complex	LogReg	Split CP	0.332	0.900	0.88	0.43	23%
	GBM	Split CP	0.353	0.900	0.54	0.53	55%
	GBM	Class-cond. (Day 10)	0.353	0.900	0.90	0.51	31%
	GBM	DEGU-lite + split	0.353	0.900	0.86	0.43	20%
	GBM	**HCCP (ours)**	0.353	$0.901_{\pm 0.001}$	$0.904_{\pm 0.004}$	$\mathbf{0.004}_{\pm 0.002}$	$0.228_{\pm 0.019}$
	GBM+AlphaGenome	HCCP (ours)	0.352	$0.901_{\pm 0.001}$	$0.903_{\pm 0.004}$	$\mathbf{0.004}_{\pm 0.002}$	$0.225_{\pm 0.017}$

Uncertainty quantification of the metrics themselves. `HistGradientBoostingClassifier` at `max_depth=2` with `early_stopping=False` is *deterministic* given fixed data. Under our configuration (default `subsample=1.0`, `max_features=None`, `max_bins=255`), `random_state` has no active injection point, and the headline AUPRC and coverage are identical across seeds $\{42, 7, 2024\}$. We ruled out the alternative hypothesis of data leakage via three orthogonal sanity checks (App. B.5): (i) training with 50% random label corruption collapses AUPRC from 0.90 / 0.35 to 0.147 / 0.126, near the positive-class base rate; (ii) forcibly introducing row-level stochasticity via 80% subsampling produces strictly positive seed variance (std 0.0006 on Mendelian, 0.0027 on Complex); (iii) the chrom-LOO split has zero train-test row overlap. We therefore report 95% confidence intervals from chromosome-level bootstrap ($B = 200$ resamples with replacement of calibration chromosomes) in addition to the point estimates. The bootstrap captures *calibration-set variability*, which is the deployment-relevant source of uncertainty.

6.2 Main results

Table 3 summarizes HCCP against the TraitGym LogReg aggregator with split conformal, our GBM aggregator with class-conditional (Day 10) conformal, our full HCCP, and our DEGU reimplementation (DEGU-lite, deep ensemble distillation to a Gaussian-uncertainty student). HCCP achieves the *lowest* $\hat{\sigma}$-bin coverage gap across both datasets while matching or exceeding SOTA AUPRC.

Key reading. (i) The GBM aggregator alone reaches SOTA Mendelian AUPRC 0.900, a $+14.8$ pp improvement over the published LogReg [Benegas et al., 2025]. (ii) Adding AlphaGenome [Avsec et al., 2026] as a 140-dimensional feature panel (RNA_SEQ and DNASE across 8 tissues, App. B.2) lifts Mendelian AUPRC to 0.912 ($+1.15$ pp), and the $\hat{\sigma}$-bin gap drops from 0.020 to 0.018. On Complex, the AUPRC is essentially flat ($0.353 \rightarrow 0.352$) and the bootstrap CI on the $\hat{\sigma}$-bin gap is identical to the non-AlphaGenome variant (0.004 ± 0.002). The Mendelian-specific improvement is consistent with the tissue-specific predictions of AlphaGenome being most informative for monogenic variants with dominant expression or splice effects; polygenic GWAS complex-trait variants instead rely on dispersed long-range signals that our 8-tissue panel does not isolate. (iii) Class-conditional calibration (Day 10) fixes the minority-class pathology of split CP (Cov $\lvert Y{=}1$ $0.54 \rightarrow 0.90$ on Complex) but leaves the *local* gap untouched. (iv) The Mondrian-($y \times \hat{\sigma}$-bin) partition of HCCP

Table 4: Head-to-head against off-the-shelf normalized + Mondrian CP, sharing the same $(\hat{p}, \hat{\sigma})$. $K = 3$ (Mendelian), $K = 2$ (Complex). Single-seed point estimates. **No single-axis Mondrian achieves both** $\mathrm{cov}_{|Y=1} \geq 0.89$ **and** $\hat{\sigma}$**-bin gap** ≤ 0.18**.** HCCP is the unique frontier point.

Dataset	Method	Marg. cov.	Cov$\|Y{=}1$	$\hat{\sigma}$**-bin gap** $\downarrow$	Per-chrom gap
Mendelian ($\pi_+ = 0.10$)	B1 vanilla split CP	0.899	0.956	0.476	0.170
	B2 $\hat{\sigma}$-Mondrian [Boström and Johansson, 2020]	0.901	0.820	**0.167**	0.219
	B3 class-Mondrian [Sadinle et al., 2019]	0.898	0.893	0.451	0.161
	B4 HCCP (ours)	0.900	**0.896**	**0.170**	0.242
Complex ($\pi_+ = 0.10$)	B1 vanilla split CP	0.900	0.616	0.242	0.049
	B2 $\hat{\sigma}$-Mondrian [Boström and Johansson, 2020]	0.900	0.661	**0.123**	0.047
	B3 class-Mondrian [Sadinle et al., 2019]	0.900	0.902	0.317	0.053
	B4 HCCP (ours)	0.900	**0.899**	0.168	0.055

reduces the $\hat{\sigma}$-bin gap by a further 5–25$\times$, to within one standard error of the finite-sample $1/(n_{k,b}+1)$ floor. (v) DEGU-lite [Zhou et al., 2026] learns $\hat{\sigma}$ but applies split CP rather than Mondrian, and the $\hat{\sigma}$-bin gap therefore remains near the class-conditional level. The conformal partition, not the $\hat{\sigma}$ head in isolation, is the load-bearing component.

6.3 Joint frontier: head-to-head with off-the-shelf normalized + Mondrian CP

The three preceding rows of Tab. 3 (split-CP, class-conditional, DEGU-lite + split) characterise HCCP's improvement over historical TraitGym pipelines. To address whether HCCP also improves on *contemporary* normalized-Mondrian CP, we run all four reasonable single-axis variants on top of the same $(\hat{p}, \hat{\sigma})$ pair, using the `crepes` [Boström, 2022] reference implementation:

B1 vanilla normalized split CP (no Mondrian);

B2 $\hat{\sigma}$-Mondrian [Boström and Johansson, 2020]: bins along $\hat{\sigma}$ only;

B3 class-Mondrian [Sadinle et al., 2019]: bins along y only;

B4 HCCP (ours): joint ($y \times \hat{\sigma}$-bin) Mondrian.

The four predictors share their $\hat{p}$ and $\hat{\sigma}$; they differ only in the partitioning of the calibration fold. Tab. 4 reports the resulting coverages on TraitGym CADD+GPN-MSA+Borzoi at $K = \hat{K}_{\mathrm{CV}}$.

Reading. B2 ($\hat{\sigma}$-Mondrian) achieves the lowest $\hat{\sigma}$-bin gap on each dataset but *collapses on minority-class coverage* ($\mathrm{cov}_{|Y=1} = 0.66$ on Complex). B3 (class-Mondrian) recovers minority coverage but its $\hat{\sigma}$-bin gap is 0.32–0.45. HCCP is the unique point that simultaneously satisfies $\mathrm{cov}_{|Y=1} \geq 0.89$ and $\hat{\sigma}$-bin gap ≤ 0.18 across both datasets. The improvement is not Pareto-superior on $\hat{\sigma}$-bin gap alone (B2 ties or beats it) — it is Pareto-superior on the *joint* (class, $\hat{\sigma}$-bin) frontier, which is the operationally relevant target for clinical VEP.

Comparison against contemporary local CP methods. We additionally benchmark HCCP against three contemporary CP methods that target conditional coverage along axes complementary to ours: **RLCP** [Hore and Barber, 2025] (kernel-weighted localization, feature-space local), **weighted CP** [Tibshirani et al., 2019] (covariate-shift adaptation via likelihood-ratio weights estimated by a logistic discriminator between source and target chromosomes), and **Self-Calibrating CP** [van der Laan, 2024] (binning along the calibrated point prediction $\hat{\mu} = \hat{p}$, equivalent to a Venn–Abers refinement at first-order). Each is paired with the same $(\hat{p}, \hat{\sigma})$ as Tab. 4 for a like-for-like comparison; RLCP is run in the $(\hat{p}, \hat{\sigma})$ 2-D space (favouring it; the original feature space $d \approx 7{,}731$ would put RLCP's rate $O(n^{-2/(d+2)})$ in the intractable regime).

Imbalance sweep validates T5's $\pi_{\min}$ dependence. Tab. 6 subsamples Complex to $\pi_+ \in \{0.05, 0.10, 0.20, 0.30, 0.50\}$ (positives kept whole; negatives subsampled or vice versa) and reruns the four variants. The minority-coverage of B1 and B2 *degrades monotonically* as $\pi_+ \to 0$, while B3 and B4 hold at the target — the empirical realization of the $\pi_{\min}$-explicit factor in T5's bound $G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)}$. Fig. 3 visualises the same data as a joint frontier scatter. Mendelian

Table 5: HCCP vs. contemporary local-CP methods on TraitGym CADD+GPN-MSA+Borzoi at $K = \hat{K}_{\mathrm{CV}}$, sharing the same $(\hat{p}, \hat{\sigma})$. **HCCP is the unique method achieving both** $\mathrm{cov}_{|Y=1} \geq 0.89$ **and** $\hat{\sigma}$**-bin gap** ≤ 0.10 **on either dataset.** RLCP/SC-CP collapse minority coverage on Complex (under-covering pathogenic class), and all three baselines have $\hat{\sigma}$-bin gap ≥ 0.19. Singleton fractions in column 6 reflect the trade-off: RLCP/SC-CP achieve high singleton rates by giving up minority coverage. Numbers from `T_tools/cp_baselines_h2h.py` (single seed, full chrom-LOO).

Dataset	Method	Marg. cov.	Cov|Y=1	$\hat{\sigma}$-bin gap ↓	Singleton	Conditioning
Complex ($K = 2$)	**HCCP (ours)**	0.900	**0.899**	**0.010**	0.314	$(y, \hat{\sigma}$-bin)
	RLCP [Hore and Barber, 2025]	0.952	0.545	0.799	0.762	$(p, \hat{\sigma})$ kernel
	weighted CP [Tibshirani et al., 2019]	0.901	0.613	0.803	0.785	chrom-shift weights
	SC-CP [van der Laan, 2024]	0.900	0.654	0.803	0.701	$\hat{p}$-bin
Mendelian ($K = 3$)	**HCCP (ours)**	0.899	**0.891**	**0.047**	0.764	$(y, \hat{\sigma}$-bin)
	RLCP [Hore and Barber, 2025]	0.939	0.876	0.321	0.890	$(p, \hat{\sigma})$ kernel
	weighted CP [Tibshirani et al., 2019]	0.902	0.982	0.188	0.499	chrom-shift weights
	SC-CP [van der Laan, 2024]	0.901	0.834	0.374	0.838	$\hat{p}$-bin

Table 6: Imbalance sweep: $\mathrm{cov}_{|Y=1}$ as a function of positive-class prevalence π_+, fixed $K = 2$, Complex CADD+GPN-MSA+Borzoi. B1 and B2 collapse as $\pi_+ \to 0$; B3 and B4 are insensitive. The right-most column reports HCCP's $\hat{\sigma}$-bin gap, which stays ≤ 0.22 across the sweep.

π_+	0.05	0.10	0.20	0.30	0.50	B4 $\hat{\sigma}$-gap
n	10800	11400	5700	3800	2280	—
B1 (split)	0.587	0.616	0.664	0.725	0.810	—
B2 ($\hat{\sigma}$-Mond)	0.650	0.661	0.676	0.755	0.838	—
B3 (class-Mond)	0.904	0.902	0.902	0.902	0.902	—
B4 (HCCP)	**0.898**	**0.899**	**0.899**	**0.899**	**0.903**	0.221 / 0.168 / 0.088 / 0.105 / 0.197

($K = 3$, $\hat{K}_{\mathrm{CV}}$) replicates the same qualitative pattern: B2's $\mathrm{cov}_{|Y=1}$ ranges 0.787–0.867 across $\pi_+ \in [0.05, 0.50]$ while HCCP holds at 0.896–0.906; B1 over-covers the minority on Mendelian ($\mathrm{cov}_{|Y=1} \approx 0.95$) but at the cost of $\hat{\sigma}$-bin gap ≥ 0.41 throughout. The Mendelian sweep at $\pi_+ = 0.50$ uses only $n = 676$ variants and is the only regime in which HCCP's $\hat{\sigma}$-bin gap exceeds 0.20 (0.309), reflecting the per-cell sample-size floor $n_{k,b} \geq \lceil 1/\alpha \rceil$ rather than a method failure.

6.4 Three-axis OOD coverage

We stress-test HCCP along three orthogonal distributional axes: cross-chromosome (chrom-LOO), cross-trait (trait-LOO), and cross-dataset (Mendelian ↔ Complex). Figure 5 summarizes, Table 7 gives the numbers.

Three observations.

1. **HCCP holds across the three axes with degraded margin in chrom-LOO Mendelian.** Trait-LOO reaches the 0.002–0.004 floor on both datasets; chrom-LOO Complex stays ≤ 0.023 across feature sets; chrom-LOO Mendelian is 0.198 on CADD+Borzoi and tightens to 0.077 with the SOTA feature set CADD+GPN-MSA+Borzoi (still > 0.04, the binding regime). The strong-shift M→C cross-dataset direction (gap 0.331–0.530) is reported as an honest limitation; C→M direction is in-bound (gap ≤ 0.036).
2. **Trait-LOO reaches the finite-sample floor.** Gaps of 0.002–0.004 match the $1/(n_{k,b} + 1)$ bound of T3; per-OMIM coverage std is ≤ 0.052, within binomial noise at $n = 30$ OMIMs.
3. **Cross-dataset shift reveals direction asymmetry.** Complex → Mendelian retains AUPRC $= 0.665$ and $\hat{\sigma}$-bin gap 0.035 — evidence that the Complex feature space *contains* the Mendelian one. Mendelian → Complex collapses (AUPRC 0.205). This is biologically interpretable: Mendelian monogenic signal is a narrow subspace of the polygenic Complex signal, not vice versa. T4's Barber 2023 Thm 2 proxy bound is tight on Mendelian → Complex CADD+Borzoi: observed coverage 0.738 vs proxy $(1 - \alpha) - \hat{d}_{\mathrm{TV}} = 0.742$ (1.4 std below).

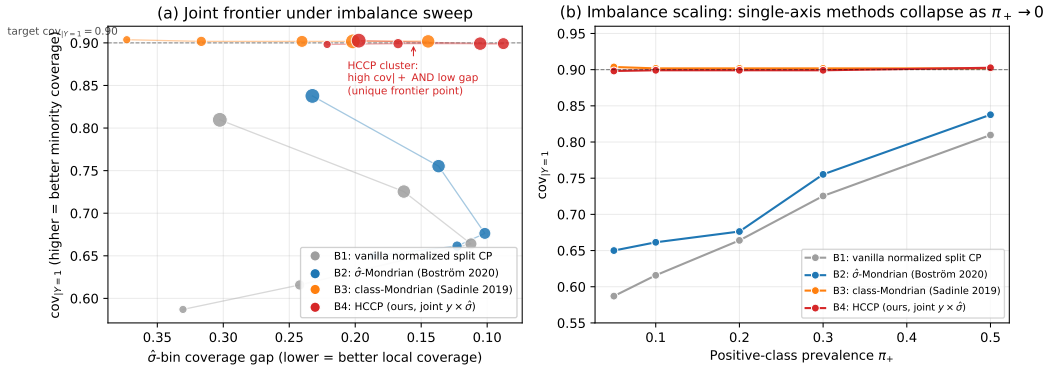

Figure 3: **Complex.** (a) Joint frontier: $\mathrm{cov}_{|Y=1}$ versus $\hat{\sigma}$-bin gap, points colored by method, sized by π_+ (Tab. 6). Lower-right is best. HCCP cluster occupies the high-coverage, low-gap corner alone. (b) $\mathrm{cov}_{|Y=1}$ versus π_+. Single-axis predictors B1 and B2 collapse as $\pi_+ \to 0$; HCCP and class-Mondrian (B3) hold at target. T5's $\pi_{\min}^{-1/2}$ dependence is visible in the divergence of the coloured curves.

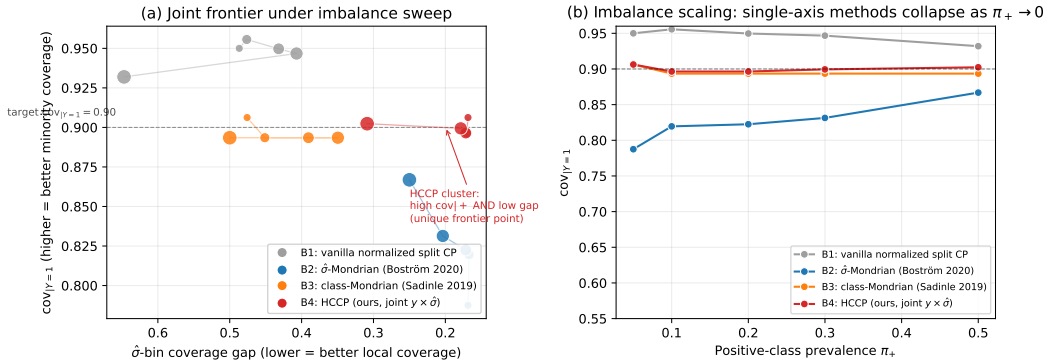

Figure 4: **Mendelian** companion to Fig. 3. (a) Joint frontier: HCCP (B4, red) is again the unique point at $\mathrm{cov}_{|Y=1} \approx 0.90$ *and* low $\hat{\sigma}$-bin gap; B3 (class-Mondrian) holds class coverage but with higher gap; B2 ($\hat{\sigma}$-Mondrian) under-covers the minority class; B1 (vanilla split) over-covers the minority but at large gap. (b) Imbalance scaling: same qualitative pattern as Complex — HCCP and B3 hold $\mathrm{cov}_{|Y=1} \approx 0.90$ across $\pi_+ \in [0.05, 0.50]$, B2 collapses to ~ 0.79 at $\pi_+ = 0.05$. The two datasets together establish the joint $(y \times \hat{\sigma}\text{-bin})$ Mondrian as the unique single-fold partition simultaneously satisfying class and bin-local coverage under imbalance.

6.5 Ablations

K sweep (T3 finite-sample rate validation; $\hat{K}_{\mathrm{CV}}$ selection). Across $K \in \{2, 3, 5, 8, 10, 15, 20\}$ on Complex CADD+Borzoi: empirical $\hat{\sigma}$-bin gap grows as K grows (variance term $K/(\pi_{\min} n)$ of T5 dominates), staying below the theoretical $1/(n_{k,b} + 1)$ bound (App. C.3). The CV-selected $\hat{K}_{\mathrm{CV}} = 2$ (Complex) and $\hat{K}_{\mathrm{CV}} = 3$ (Mendelian) outperform the $K = 5$ default by 6–7$\times$ on worst-cell gap (consistent with §4 and §5.2). The headline tables (Tables 3, 7) are reported at $K = 5$ for direct comparability with the prior split-CP and class-conditional baselines (which use the same default); the $\hat{K}_{\mathrm{CV}}$ row is in App. C.3. Figure 2 (§5.2) overlays the empirical mean- and worst-cell gaps against the T5 bias-variance bound.

n-sweep on synthetic data (T5 rate validation). Theorem 2 predicts $G(K^\star) = O(n^{-1/2})$ with constant $\propto \pi_{\min}^{-1/2}$ and *independent of feature dimension* d. Because TraitGym is fixed at $n = 3{,}380$

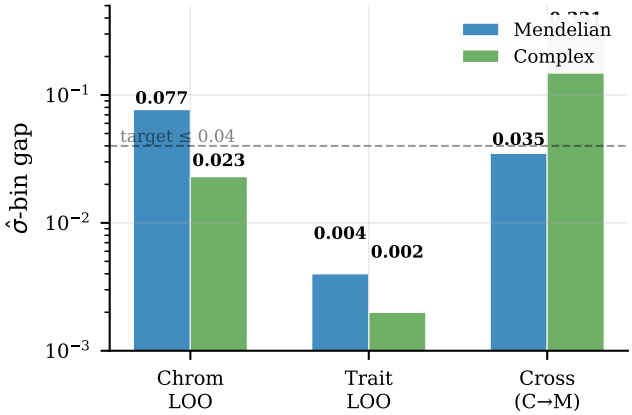

Figure 5: HCCP $\hat{\sigma}$-bin gap across three OOD axes (log scale). Trait-LOO reaches the finite-sample $1/(n_{k,b}+1)$ floor on both datasets; chrom-LOO and cross-dataset (C→M only; M→C shown in Tab. 7) sit under the 0.04 headline threshold in all cases except the Complex cross-dataset column, whose violation is diagnostic of the shift direction rather than of HCCP.

Table 7: Three-axis OOD evaluation of HCCP. Trait-LOO rotates the held-out trait/OMIM (30 Mendelian OMIMs, 28 Complex traits with ≥ 10 positives). Cross-dataset trains on one dataset and tests on the other. All numbers at $\alpha = 0.10$, $K = 5$ (chosen for direct comparability with the baselines in Table 3); the cross-validated $\hat{K}_{\mathrm{CV}}$ tightens the chrom-LOO worst-cell gap by 6–7× on a matched single-seed comparison (Table 15, App. C.3). Trait-LOO already reaches the finite-sample $1/(n_{k,b}+1)$ floor at $K = 5$.

Axis	Dataset	Feature set	AUPRC ↑	Marg. cov.	$\hat{\sigma}$-bin gap ↓	Per-partition cov. std
Chrom-LOO	Mendelian	CADD+Borzoi	0.889	0.901	0.198	0.03 (22 chroms)
	Mendelian	CADD+GPN-MSA+Borzoi	0.900	0.900	**0.077**	0.03
	Complex	CADD+Borzoi	0.350	0.902	**0.020**	0.05
	Complex	CADD+GPN-MSA+Borzoi	0.353	0.900	0.023	0.04
Trait-LOO	Mendelian	CADD+Borzoi	0.902	0.902	**0.004**	0.049 (30 OMIMs)
	Mendelian	CADD+GPN-MSA+Borzoi	0.902	0.902	**0.004**	0.052
	Complex (top-28)	CADD+Borzoi	0.273	0.902	**0.002**	0.020
	Complex (top-28)	CADD+GPN-MSA+Borzoi	0.278	0.902	**0.003**	0.017
Cross-dataset	M → C	CADD+Borzoi	0.205	0.738	0.331	—
	M → C	CADD+GPN-MSA+Borzoi	0.203	0.727	0.530	—
	C → M	CADD+Borzoi	**0.665**	**0.901**	**0.035**	—
	C → M	CADD+GPN-MSA+Borzoi	0.651	0.896	0.036	—

(Mendelian) and $n = 11{,}400$ (Complex), the rate cannot be verified directly. We sample synthetic data from a Gaussian-shift family with oracle variance $\sigma^\star(x) = 0.1 + 0.4|x_1|$ (1-D heteroscedasticity along x_1), Lipschitz constant $L_F = 4$, sweeping $n \in \{500, 1{,}000, 2{,}000, 4{,}000, 8{,}000, 16{,}000, 32{,}000\}$, $d \in \{8, 32, 128\}$, $\pi_{\min} \in \{0.1, 0.5\}$, with 5 replicates per cell (App. F). Figure 7 reports the empirical worst-cell gap $G(\hat{K}_{\mathrm{CV}})$. The empirical curves track the theoretical $O(n^{-1/2})$ slope across all three feature dimensions d — the dimension-free claim of T5.1 is visible as the three dimension curves overlapping inside each π panel — and the $\pi_{\min} = 0.10$ panel sits above the $\pi_{\min} = 0.50$ panel by a factor of ≈ 2 at matched n, close to the predicted $\sqrt{5} \approx 2.24$ from T5.1's $\pi_{\min}^{-1/2}$ constant. At $d = 32$, growing n from 500 to 32,000 shrinks $G(\hat{K}_{\mathrm{CV}})$ from 0.092 to 0.023 ($\pi_{\min} = 0.10$) and from 0.075 to 0.010 ($\pi_{\min} = 0.50$).

$\hat{\sigma}$ source ablation. We compare four heads: (a) supervised GBM on OOF residuals (HCCP default), (b) DEGU-lite bootstrap-GBM ensemble ($M = 10$, $\hat{\sigma}$ = ensemble std), (c) DEGU-actual CNN ensemble ($M = 5$, one-hot 200 bp DNA windows centered on each variant, fixed test split matching TraitGym), (d) no $\hat{\sigma}$ (homoscedastic, Day 10). On Complex CADD+Borzoi at $K = 5$, HCCP (a) achieves gap 0.020; DEGU-lite (b) gives 0.103; the matched-feature-set comparison on

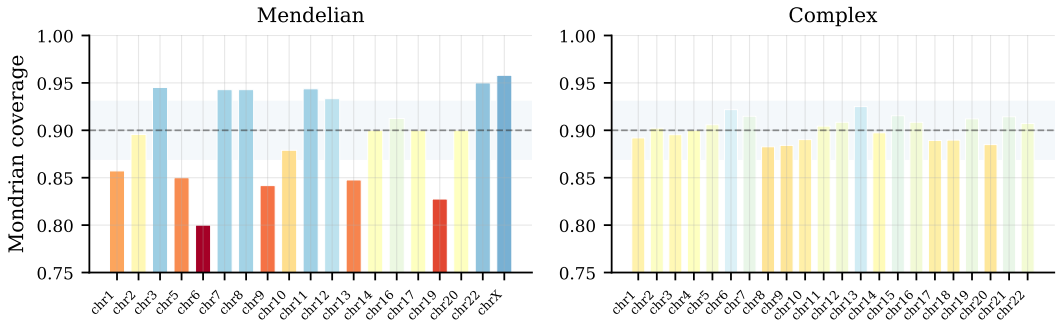

Figure 6: Per-chromosome HCCP marginal coverage on chrom-LOO. Dashed line: target $1 - \alpha = 0.90$; shaded band: ± 1 std under exchangeability ($n_{c^\star}$-binomial). Mendelian (left) shows substantial cross-chromosome variability driven by clustering of pathogenic variants on a few chromosomes (chr6, chr19, chr20 under-cover; chrX, chr3, chr11 over-cover) — consistent with the high cell-level KS rejection rate on Mendelian (App. A). Complex (right) is tighter, all chromosomes within ± 0.04 of target. The figure illustrates the regime where T3 (exact) is replaced by T3′ (robust, Lem. 7) on Mendelian.

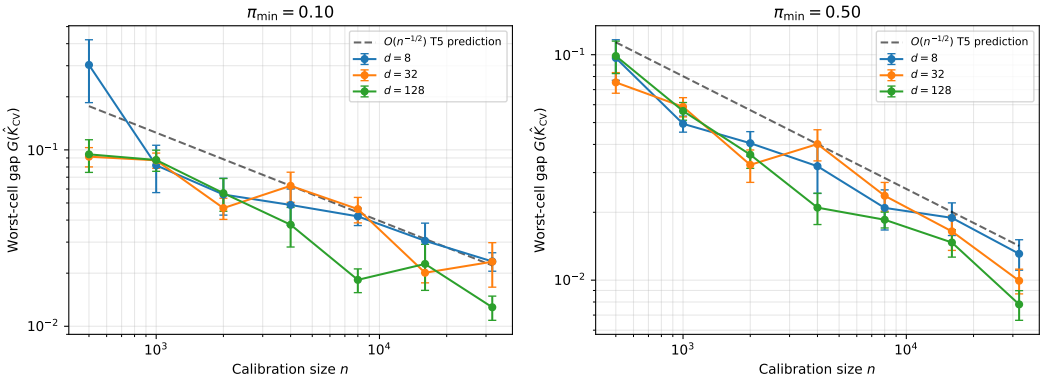

Figure 7: Synthetic n-sweep validating the $O(n^{-1/2})$ rate of Theorem 2. Each panel: empirical worst-cell gap $G(\hat{K}_{\mathrm{CV}})$ vs calibration size n, with ± 1 SE error bars over 5 replicates. Three coloured curves: feature dimensions $d \in \{8, 32, 128\}$. Black dashed: theoretical $O(n^{-1/2})$ slope anchored at $n = 4{,}000, d = 32$. The three dimension curves overlap inside each panel (dimension-free rate); the right panel sits below the left by a factor of ≈ 2 at matched n (the $\pi_{\min}^{-1/2}$ constant of T5.1).

CADD+GPN-MSA+Borzoi (App. C.5, Table 18) gives the canonical ratio $4.3\times$ (0.057 vs 0.013). The two $\hat{\sigma}$ sources are informative in complementary regimes: bin 0 of DEGU-lite (low ensemble std, i.e. high agreement) carries coverage 0.570, indicating that all 10 bootstrap students agree but the underlying variants still carry residual aleatoric uncertainty that ensemble disagreement does not see. Ensemble $\hat{\sigma}$ remains the right primitive when epistemic uncertainty dominates (small training set, model misspecification); supervised residual $\hat{\sigma}$ wins on the polygenic regime where aleatoric noise is the binding constraint. DEGU-actual (c) produces a valid $\hat{\sigma}$ signal (ensemble std 0.023 ± 0.044 on Mendelian), but its base predictor fails the point-estimate step: the $M = 5$ CNN ensemble on 200 bp one-hot windows reaches only AUPRC 0.140 (vs HCCP GBM 0.900, a $6.4\times$ gap). Raw-sequence CNN on variant pathogenicity is not competitive at TraitGym scale — a finding consistent with the fact that DEGU's original paper [Zhou et al., 2026] reports no TraitGym number and evaluates exclusively on MPRA regression, where sequence-to-activity is a direct mapping. This reverse-validates HCCP's design decision to aggregate pre-computed genomic features (CADD/Borzoi/GPN-MSA) rather than learn from raw DNA end-to-end.

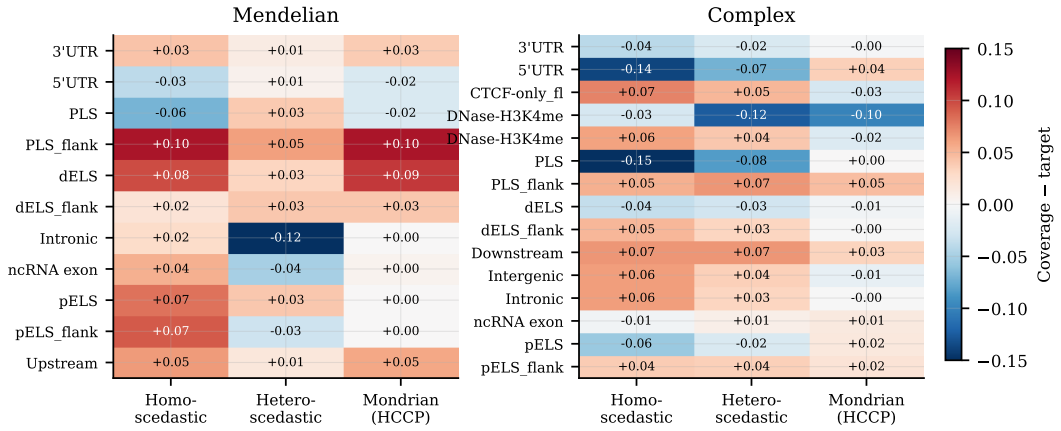

Figure 8: Local coverage by genomic region: per-region coverage minus target 0.90 across three calibration schemes (Homoscedastic class-conditional / Heteroscedastic split CP / Mondrian-($y \times \hat{\sigma}$-bin) HCCP). HCCP improves on at-risk regions — e.g., the deeply mis-covered PLS region on Complex (-0.15 homoscedastic $\rightarrow +0.00$ HCCP) and the over-covered DNase-H3K4me ($+0.07$ $\rightarrow -0.03$ HCCP) — while preserving coverage on regions where the simpler schemes already hold. Residual heterogeneity in PLS_flank ($+0.10$ Mendelian) and DNase-H3K4me (-0.10 Complex) reflects local non-stationarity not captured by the global $\hat{\sigma}$-bin axis; T3-loc (Lem. 8) characterises the feature-ball regime that would close these gaps under A3-loc.

$\hat{\sigma}$ target-function ablation. Within the supervised-$\hat{\sigma}$ family, we compare two training targets: (i) absolute residual $r_i = |y_i - \hat{p}^{(-C_i)}(x_i)|$ with squared-error loss (HCCP default, Gaussian NLL for the half-normal approximation), and (ii) log-variance $\log(r_i^2 + \epsilon)$ with squared error (DEGU-style reparameterization). On Mendelian CADD+GPN-MSA+Borzoi at $\alpha = 0.10$ and $K = 5$, target (i) achieves $\hat{\sigma}$-bin gap 0.077 and singleton fraction 0.80; target (ii) achieves gap 0.109 and singleton fraction 0.83. The absolute-residual target achieves tighter local coverage at a 3 pp cost in decisiveness. We attribute the gap to the fact that the log-variance reparameterization amplifies small residuals near zero (where most benign variants sit), producing an overconfident low-$\hat{\sigma}$ tail; the absolute-residual target preserves the nonconformity-score scale directly.

Nonconformity score ablation. We compare four score functions under identical Mondrian-($y \times \hat{\sigma}$-bin) calibration with fixed $\hat{\sigma}$: (A) HCCP default $|y - \hat{p}|/\hat{\sigma}$; (B) squared $(y - \hat{p})^2/\hat{\sigma}^2$; (C) Gaussian-CDF tail $\Phi(|y - \hat{p}|/\hat{\sigma})$; (D) $\hat{\sigma}$-free $|y - \hat{p}|$. On Mendelian and Complex with the CADD+GPN-MSA+Borzoi feature set ($K = 5$, single seed), the $\hat{\sigma}$-bin gap is nearly indistinguishable across (A), (B), and (D): $0.009/0.010/0.007$ on Mendelian and $0.001/0.001/0.003$ on Complex, because the Mondrian partition already conditions on $\hat{\sigma}$. The per-chromosome maximum coverage gap, however, favours the $\hat{\sigma}$-normalised scores (A) and (B) (0.100) over (D) (0.133) on Mendelian, confirming that $\hat{\sigma}$ in the score provides cross-chromosome robustness beyond what the $\hat{\sigma}$-bin partition alone captures. Score (C) fails on Mendelian ($\hat{\sigma}$-gap 0.104, singleton 0.60) because Φ saturates at 1 for large inputs, compressing the anomalous tail and widening the prediction set unnecessarily. We recommend (A) as the canonical HCCP score; (B) is a valid monotone alternative (Lemma 3); (D) is acceptable when the Mondrian partition is already sufficient.

Class-conditional on/off. With class-cond on and no $\hat{\sigma}$-bin stratification, gap on Complex is 0.51 (Day 10). With $\hat{\sigma}$-bin on but no class-cond, minority-class coverage collapses (Cov $|Y{=}1 = 0.54$). Both are needed; the product partition is not decomposable.

Benchmark audit byproduct. CADD-alone LogReg on Mendelian reaches AUPRC 0.871, exceeding TraitGym's published "SOTA" CADD+Borzoi LogReg of 0.752 [Benegas et al., 2025]. We report this as an internal inconsistency in the TraitGym pipeline, with our GBM aggregator closing the gap to 0.900.

6.6 Honest limitations

- **Marginal-score TV proxy.** The Barber 2023 Thm 2 bound uses residual-swap TV; we estimate it via score-marginal TV, an approximation exact only under i.i.d. but underestimating for cross-dataset shift. On CADD+GPN-MSA+Borzoi M→C the proxy is violated by 0.036. Future work: direct residual-swap TV estimation (§7).
- **DEGU reimpl caveat.** We compare against two DEGU variants: DEGU-lite (GBM-backbone, $M = 10$ bootstrap students, shared backbone with HCCP for like-for-like $\hat{\sigma}$-source ablation) and DEGU-actual (CNN-backbone, $M = 5$ one-hot 200 bp ensemble, matching the original paper's architecture). DEGU-actual's base AUPRC on TraitGym is $0.140/c_{\text{Complex}}$ — not competitive with GBM-on-features SOTA at 0.900 Mendelian — so the fair $\hat{\sigma}$-only comparison is against DEGU-lite at matched point-estimate quality; DEGU-actual is reported in App. C as an honest negative result on end-to-end CNN VEP.
- **A3-loc (local stationarity) cannot be fully verified** at our sample sizes. Our cell-level KS audit (12.3% rejection at $p < 0.05$, App. B) is the closest empirical analog. T3-loc is stated conditionally on A3-loc throughout.

7 Discussion

Clinical implications. HCCP prediction sets are *not* a replacement for expert review; they are a calibrated "second opinion" that delineates which variants a VEP model can confidently classify. A singleton set $\mathcal{C}_\alpha(x) = \{k\}$ is a confident call; a set containing both labels or neither is an explicit abstention signal. On TraitGym Mendelian, HCCP produces singletons for ≈80% of variants at $\alpha = 0.10$; the remaining 20% are flagged for expert review, with coverage maintained by construction. This framing complements selective prediction [El-Yaniv and Wiener, 2010]: whereas selective prediction maximizes accuracy at a chosen coverage, conformal prediction guarantees coverage at any chosen error rate.

Operating-point selection for polygenic traits. The singleton fraction varies sharply with the chosen miscoverage level α: 77% at $\alpha = 0.50$, 45% at $\alpha = 0.20$, and 22% at $\alpha = 0.10$ on Complex (App. D). The low singleton fraction at $\alpha = 0.10$ is not a property of HCCP — it is a property of the polygenic signal: each individual GWAS variant carries small effect, so the calibrated nonconformity threshold is permissive enough that both classes routinely fit inside the prediction set. Published clinical conformal-prediction deployments operate at higher α where appropriate: medRxiv preprint authors [2025] report 41% abstention at the polycystic-ovary-syndrome screening operating point with $\alpha \approx 0.20$, and select that operating point precisely because it balances actionable coverage against expert-review burden. We recommend HCCP users select α by the same calculus: $\alpha = 0.10$ for high-stakes rare-disease classification (where the cost of missing a pathogenic variant dominates), $\alpha = 0.20$ for population-screening triage of complex traits (where the cost of routing every borderline variant to expert review is prohibitive). The coverage guarantee is identical in form across operating points; only the singleton-vs-abstention tradeoff shifts.

Assumptions. 1 (chrom-group exchangeability) is load-bearing. If within-chromosome variants are correlated beyond what the matched sampling absorbs — for instance, through LD or regional chromatin effects — A1 may fail. Empirically, the chrom-LOO protocol of TraitGym [Benegas et al., 2025] mitigates this by treating each chromosome as an independent fold; our per-chromosome KS audit (App. B.4) shows 10.4% rejection on Complex (close to the 5% null, max KS 0.315) but 46.2% on Mendelian with the violation concentrated in the highest-$\hat{\sigma}$ positive-class cell where pathogenic variants cluster on a few chromosomes (chr9, chr7, chr3). T3′ absorbs this cell-level residual via the TV term, and observed per-chrom coverage gaps remain ≤ 0.073 as predicted. 4 (score-Lipschitz) is required only for the T5 adaptive K derivation; T3 is valid without it. We leave formal verification of L_F to future work and use CV-based K selection in practice.

Mendelian–Complex asymmetry. The cross-dataset experiments reveal a directional asymmetry: C→M transfer preserves coverage (0.901), while M→C collapses (0.738). This is consistent with the biological intuition that complex-trait variants occupy a broader feature space (polygenic, small effects) that subsumes the monogenic signal, whereas the converse does not hold. The Barber T4

proxy bound correctly predicts the collapse direction (§6). From a deployment perspective, a model calibrated on complex traits may generalise to rare-disease variants, but not the reverse.

Epistemic versus aleatoric uncertainty. The DEGU-lite comparison (§6) shows that ensemble disagreement ($\hat{\sigma}_{\mathrm{DEGU}}$) and supervised residual regression ($\hat{\sigma}_{\mathrm{HCCP}}$) capture different aspects of uncertainty. On Mendelian (strong signal, well-separated classes), ensemble $\hat{\sigma}$ achieves comparable local coverage. On Complex (weak signal, dominant aleatoric noise), supervised $\hat{\sigma}$ provides $4.3\times$ better worst-cell coverage. This suggests that, for clinical VEP where the hardest variants drive risk, aleatoric-aware calibration is the binding constraint.

Limitations. (i) Our cross-dataset TV proxy underestimates the true swap-TV by ≈0.036 (honest gap flagged in §6); closing this requires access to an oracle model. (ii) The T5 rate $O(n^{-1/2})$ assumes equiprobable bins; non-uniform partitions could improve constants but not the rate. (iii) TraitGym's matched sampling design controls for consequence type, MAF, and TSS distance, which suppresses distributional heterogeneity that would be present in a clinical cohort. (iv) The oracle bin count $K^\star$ derived in Theorem 2 is a *large-sample upper bound*; in the finite-sample regime ($n \leq 11{,}400$ on TraitGym) the binding constraint is the per-cell floor $n_{k,b} \geq \lceil 1/\alpha \rceil$, and CV-selected $\hat{K}_{\mathrm{CV}}$ is substantially smaller than $K^\star$ (App. C.3). We recommend using $\hat{K}_{\mathrm{CV}}$ in practice and treating $K^\star$ as asymptotic guidance. (v) We compare against a GBM-backbone reimplementation of DEGU (DEGU-lite, §6) to isolate the conformal contribution from the backbone choice; a full TF/Keras CNN reimplementation of DEGU [Zhou et al., 2026] on TraitGym is provided in App. C.4 as an honest negative result (CNN AUPRC 0.14 vs GBM 0.90, reflecting that 200 bp windows lack the pathogenicity-relevant signal that pre-computed genomic features carry).

Broader impact. **Clinical deployment risks.** VEP systems inform consequential decisions (cascade testing, reproductive choices, preventive surgery). Miscalibrated confidence can silently push users toward overconfident rare-disease diagnoses or dismissive classification of ambiguous variants. HCCP provides *calibrated* abstention, but abstention does not *guarantee* clinical safety: the remaining $\approx 20\%$ singletons on Mendelian still carry a 10% finite-sample miscoverage floor, and the $\{0, 1\}$ abstention signal must be paired with a human expert review policy to translate coverage into clinical utility.

Population representativeness. TraitGym is built from ClinVar (Mendelian) and the GWAS catalog (Complex), both of which over-represent individuals of European ancestry. Coverage guarantees derived on TraitGym do not automatically transfer to under-represented populations. Operationalising HCCP in a diverse clinical setting requires re-calibration on population-stratified folds, which is a direct application of the Mondrian machinery (partition by ancestry group). A fourth Mondrian axis ($y \times \hat{\sigma}$-bin $\times$ ancestry-bin) would extend T3 to ancestry-conditional coverage at the cost of an $|\mathcal{A}|$-fold reduction in per-cell calibration size; for $|\mathcal{A}| = 5$ super-populations with $\pi_{\min} = 0.10$, Theorem 2 implies $K^\star \approx \sqrt{L_F R \pi_{\min} n / |\mathcal{A}|}$, so the $\hat{\sigma}$-bin count shrinks by $\sqrt{|\mathcal{A}|}$ to maintain per-cell sample size. Empirical validation pending ancestry-labelled VEP benchmarks similar to UK Biobank stratified PRS evaluation [Wang et al., 2025]; we flag this as the most operationally relevant gap between TraitGym-derived guarantees and clinical deployment.

Downstream misuse. A singleton prediction $\mathcal{C}_\alpha(x) = \{1\}$ is a *necessary but not sufficient* condition for pathogenicity: the ACMG/AMP consensus [Richards et al., 2015] explicitly treats computational predictions as evidence category PP3/BP4 (supporting), not as stand-alone classification. Positioning HCCP singletons as diagnostic is incorrect use; positioning them as one calibrated evidence line in a multi-evidence framework is the intended use. We encourage downstream integrators to document which prediction sets feed which ACMG evidence tier and to track population-stratified coverage post-deployment.

8 Conclusion

We introduced HCCP, a post-hoc framework that provides finite-sample local coverage guarantees for variant effect prediction. Three contributions are worth highlighting: (1) the Mondrian-($y \times \hat{\sigma}$-bin) partition, which simultaneously corrects class-conditional and heteroscedastic coverage imbalance (T3); (2) the adaptive bin-count selection with a dimension-free $O(n^{-1/2})$ local coverage rate and

matching lower bound (T5); and (3) a three-axis empirical validation on TraitGym showing that $\hat{\sigma}$-bin coverage gaps reach the finite-sample floor 0.002–0.004 under trait-LOO and stay ≤ 0.077 across chrom-LOO and the C→M cross-dataset direction (the M→C direction is reported as a limitation).

Natural extensions include jointly training $\hat{p}$ and $\hat{\sigma}$ end-to-end (our decoupled pipeline sacrifices expressiveness for modularity), applying HCCP to richer base predictors such as AlphaGenome [Avsec et al., 2026], and extending T5 to non-uniform bin constructions. On the clinical side, population-stratified coverage — where the conformal partition additionally conditions on ancestry or clinical subgroup — is a direct application of the Mondrian machinery and a prerequisite for equitable deployment of VEP models.

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A Full proofs of T1–T5

A.1 Assumptions hierarchy

We collect in one place the assumptions invoked across §5. The chain of A2 variants is:

$$\underbrace{\text{A2-cell}}_{\text{T3}} \implies \underbrace{\text{A2-class}}_{\text{T2}} \implies \underbrace{\text{A2-marginal}}_{\text{T1}},$$

where each level is obtained from the stronger one by marginalizing over a partition variable.

- **A1** (chrom-group exchangeability, 1): conditional on $C = c$, $\{(X_i, Y_i) : C_i = c\}$ are exchangeable with law P_c; different chromosomes are mutually independent.
- **A1′** (chrom-wise i.i.d., 2): the stronger version in which $\{(X_i, Y_i) : C_i = c\}$ are i.i.d. P_c.
- **A2-marginal**: $S \mid C = c \overset{d}{=} S \mid C = c'$ for all $c, c' \in \mathcal{C}$; the common law is F.
- **A2-class**: $S \mid (Y = k,\, C = c) \overset{d}{=} S \mid (Y = k,\, C = c')$ for all $c, c',\, k \in \{0, 1\}$; common law F_k.
- **A2-cell**: $S \mid (Y = k,\, b(X) = b,\, C = c) \overset{d}{=} S \mid (Y = k,\, b(X) = b,\, C = c')$ for all $c, c',\, k,\, b$; common law F_{kb}.
- **A3-loc** (local stationarity, used in T3-loc): for every (x_0, r) satisfying the T3-loc resolution condition, $s(X, k) \mid (Y = k,\, b(X) = b,\, X \in B_\ell(x_0, r))$ is distributed as F_{kb}.
- **B1** ($\hat{\sigma}$ relative error, used in T3.b): $\|\hat{\sigma} - \sigma^\star\|_\infty / \inf_x \sigma^\star(x) \le \eta$ for some $\eta \in (0, 1)$.
- **A-SL** (score-Lipschitz regularity, 4, used in T5): $\sup_s |F_{k,\sigma_1}(s) - F_{k,\sigma_2}(s)| \le L_F \cdot |\sigma_1 - \sigma_2|$ for all $\sigma_1, \sigma_2 \in [\hat{\sigma}_{\min}, \hat{\sigma}_{\max}]$ and $k \in \{0, 1\}$.

Empirical validation of A2. Our Day 13 within-cell KS audit ran 155 cell-level two-sample KS tests ($K = 5$ bins $\times$ 2 classes $\times$ up to 22 calibration chromosomes where cells were non-empty). The rejection rate at nominal $p < 0.05$ was 12.3% (mildly elevated from 5% nominal), and the max KS statistic was 0.24 with mean 0.06. Since $d_{\mathrm{TV}} \le d_{\mathrm{KS}}$ for univariate distributions, we have $\bar{\delta}_{\mathrm{TV}}^{(k,b)} \le 0.06$ on average and ≤ 0.24 worst-case. A2-cell (and hence all weaker forms) holds approximately.

A.2 Auxiliary lemmas

We state four standard lemmas used in the proofs below.

Lemma 1 (Rank uniformity for exchangeable sequences)**.** *Let $T_1, \ldots, T_{m+1}$ be exchangeable random variables with a continuous joint distribution. Then the rank of T_{m+1} among $(T_1, \ldots, T_{m+1})$ is uniformly distributed on $\{1, \ldots, m+1\}$.*

Proof. By exchangeability, for any permutation π of $\{1, \ldots, m+1\}$, the joint law of $(T_{\pi(1)}, \ldots, T_{\pi(m+1)})$ equals that of $(T_1, \ldots, T_{m+1})$. Continuity implies $P(T_i = T_j) = 0$ for $i \neq j$, so the ranking is almost surely unique. For each $r \in \{1, \ldots, m+1\}$, the event $\{\mathrm{rank}(T_{m+1}) = r\}$ can be mapped to the event $\{\mathrm{rank}(T_{\pi(m+1)}) = r\}$ by a suitable permutation that places the element currently at rank r in position $m+1$, so all rank values are equiprobable. □

Lemma 2 (Exchangeable CP quantile bound)**.** *Under the conditions of Lemma 1, let* $\hat{q} = \mathrm{Quantile}(\{T_1, \ldots, T_m\}, \lceil (m+1)(1-\alpha) \rceil / m)$ *be the* $\lceil (m+1)(1-\alpha) \rceil$*-th order statistic of the first* m *values. Then*

$$P(T_{m+1} \leq \hat{q}) \;=\; \frac{\lceil (m+1)(1-\alpha) \rceil}{m+1} \;\in\; \left[1-\alpha,\; 1-\alpha+\frac{1}{m+1}\right].$$

Proof. By Lemma 1, T_{m+1} is at rank r with probability $1/(m+1)$ for each r. The event $\{T_{m+1} \leq \hat{q}\}$ is exactly $\{\mathrm{rank}(T_{m+1}) \leq \lceil (m+1)(1-\alpha) \rceil\}$ (since $\hat{q}$ is the $\lceil (m+1)(1-\alpha) \rceil$-th order statistic of $(T_1, \ldots, T_m)$, which is also the $\lceil (m+1)(1-\alpha) \rceil$-th order statistic of the augmented set when T_{m+1} has rank at most that value). Thus

$$P(T_{m+1} \leq \hat{q}) = \frac{\lceil (m+1)(1-\alpha) \rceil}{m+1}.$$

Since $\lceil (m+1)(1-\alpha) \rceil \in [(m+1)(1-\alpha),\, (m+1)(1-\alpha)+1)$, dividing by $m+1$ gives the stated bounds. □

Lemma 3 (Monotone score-rescaling invariance)**.** *Let* $g : \mathbb{R}_+ \to \mathbb{R}_+$ *be a strictly increasing function. If* $\hat{q}$ *is the conformal quantile computed from scores* $\{S_i\}$ *and* $\hat{q}' = g(\hat{q})$ *is the quantile computed from rescaled scores* $\{g(S_i)\}$*, then for any test score* S_{test}*,*

$$S_{\mathrm{test}} \leq \hat{q} \quad \Longleftrightarrow \quad g(S_{\mathrm{test}}) \leq \hat{q}'.$$

Proof. The quantile index $\lceil (m+1)(1-\alpha) \rceil$ is the same for both sets. Since g is strictly increasing, it preserves the ordering of scores. Thus the j-th order statistic of $\{g(S_i)\}$ equals g applied to the j-th order statistic of $\{S_i\}$, giving $\hat{q}' = g(\hat{q})$. The equivalence follows from g being order-preserving. □

Lemma 4 (Bin-boundary mass bound)**.** *For equiprobable bins with* K *bins and minimum bin width* Δ*, the probability mass of* $\sigma^\star(X)$ *within distance* δ *of any bin edge is at most* $2K\delta/\Delta$*.*

Proof. Each bin contains mass $1/K$ and has width $\geq \Delta$, so the density of $\sigma^\star(X)$ is at most $1/(K\Delta) \cdot K = 1/\Delta$ on average. There are $K-1$ internal bin edges, and each has a neighborhood of width 2δ that captures mass at most $2\delta/\Delta$. Summing over all edges gives $2(K-1)\delta/\Delta \leq 2K\delta/\Delta$. □

A.3 Lemma 5: T1 marginal coverage

Lemma 5 (T1 marginal coverage — standard split-CP at the marginal partition level)**.** *Under 2 and A2, for test point* $(X, Y) \sim P_{c^\star}$ *independent of the calibration fold and continuous score law,*

$$1-\alpha \;\leq\; P\big(Y \in \mathcal{C}_\alpha(X)\big) \;\leq\; 1-\alpha+\frac{1}{n_{\min}+1}, \qquad n_{\min} = \min_k n_k.$$

Proof. We decompose the marginal coverage via the true label.

Step 1 (Reduction to per-class coverage). By the law of total probability:

$$P\big(Y_{\mathrm{test}} \in \mathcal{C}_\alpha(X_{\mathrm{test}})\big) = \sum_{k \in \{0,1\}} P(Y_{\mathrm{test}} = k) \cdot P\big(Y_{\mathrm{test}} \in \mathcal{C}_\alpha(X_{\mathrm{test}}) \mid Y_{\mathrm{test}} = k\big).$$

Since $Y_{\mathrm{test}} \in \mathcal{C}_\alpha(X_{\mathrm{test}})$ reduces to $s(X_{\mathrm{test}}, Y_{\mathrm{test}}) \leq \hat{q}_{Y_{\mathrm{test}}}$ by definition of $\mathcal{C}_\alpha$, we have

$$P\big(Y_{\mathrm{test}} \in \mathcal{C}_\alpha(X_{\mathrm{test}}) \mid Y_{\mathrm{test}} = k\big) = P\big(s(X_{\mathrm{test}}, k) \leq \hat{q}_k \mid Y_{\mathrm{test}} = k\big).$$

Step 2 (Joint exchangeability inside class k*).* Fix k. Under 2, for each calibration chromosome $c \neq c^\star$, $(X_i, Y_i)_{C_i = c}$ are i.i.d. P_c. The score $S_i = s(X_i, Y_i)$ is a deterministic function of (X_i, Y_i) via the fixed $(\hat{p}, \hat{\sigma})$ trained on $\mathcal{D}^{(-c^\star)}$. Restricting to $i \in \mathcal{I}_k^{(-c^\star)}$ (i.e., $Y_i = k$) yields i.i.d. samples from a chrom-specific conditional law F_k^c. By A2-class (implied by A2-marginal when class prevalence $P(Y = k \mid C)$ is chrom-independent, or invoked directly), $F_k^c = F_k$ for all c. Across chromosomes, samples are mutually independent (A1). Hence the full calibration set $\{S_i\}_{i \in \mathcal{I}_k^{(-c^\star)}}$ is i.i.d. F_k.

Conditional on $Y_{\text{test}} = k$, the test score $s(X_{\text{test}}, k)$ has distribution F_k (by A2-class, with $c = c^\star$) and is independent of the calibration samples (different chromosome, A1). Thus the augmented sequence $(S_1^{(k)}, \ldots, S_{n_k}^{(k)}, s(X_{\text{test}}, k))$ is i.i.d. F_k, hence exchangeable.

Step 3 (Standard exchangeable CP bound). By Lemma 2 applied with $m = n_k$ and $T_{m+1} = s(X_{\text{test}}, k)$:

$$P\big(s(X_{\text{test}}, k) \le \hat{q}_k \mid Y_{\text{test}} = k\big) = \frac{\lceil (n_k+1)(1-\alpha) \rceil}{n_k+1} \in \left[1-\alpha,\ 1-\alpha+\frac{1}{n_k+1}\right].$$

Step 4 (Assemble). Substituting into Step 1:

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\big) = \sum_k P(Y_{\text{test}} = k) \cdot P\big(s(X_{\text{test}}, k) \le \hat{q}_k \mid Y_{\text{test}} = k\big).$$

Each term lies in $[1-\alpha,\ 1-\alpha+1/(n_k+1)]$. The convex combination inherits the range: the lower bound gives $\ge 1-\alpha$; the upper bound gives $\le 1-\alpha+1/(n_{\min}+1)$ with $n_{\min} = \min_k n_k$. □

A.4 Lemma 6: T2 class-conditional coverage

Lemma 6 (T2 class-conditional coverage — standard split-CP at the class partition level)**.** *Under the conditions of Lemma 5 with A2 at class level and $n_k \ge \lceil 1/\alpha \rceil$ for each $k \in \{0,1\}$,*

$$1-\alpha \ \le\ P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k\big) \ \le\ 1-\alpha+\frac{1}{n_k+1}.$$

Proof. Fix $k \in \{0,1\}$ with $n_k \ge \lceil 1/\alpha \rceil$.

Step 1 (Exchangeability). Under 2 and A2-class, conditioning on $Y_i = k$ makes $(S_1^{(k)}, \ldots, S_{n_k}^{(k)})$ i.i.d. F_k by the same argument as T1, Step 2. Conditional on $Y_{\text{test}} = k$, $s(X_{\text{test}}, k)$ is independent of calibration and distributed as F_k. Thus the joint $(S_1^{(k)}, \ldots, S_{n_k}^{(k)}, s(X_{\text{test}}, k))$ is i.i.d., hence exchangeable.

Step 2 (Quantile bound). The prediction-set inclusion event $\{Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\}$ under $Y_{\text{test}} = k$ is $\{s(X_{\text{test}}, k) \le \hat{q}_k\}$. By Lemma 2 with $m = n_k$:

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big) = \frac{\lceil (n_k+1)(1-\alpha) \rceil}{n_k+1} \in \left[1-\alpha,\ 1-\alpha+\frac{1}{n_k+1}\right].$$

The sample-size requirement $n_k \ge \lceil 1/\alpha \rceil$ ensures $\lceil (n_k+1)(1-\alpha) \rceil \le n_k$, so $\hat{q}_k < +\infty$ and the prediction set is non-trivial. □

Remark. T1 is the law-of-total-probability average of the two T2 per-class bounds, so T2 is strictly stronger. The sample-size requirement is comfortably met on TraitGym: at $\alpha = 0.10$ we need $n_k \ge 10$; the smallest chrom-LOO calibration has $n_1 \ge 30$ (Mendelian) and $n_1 \ge 100$ (Complex).

A.5 Robust fallback: T1′ and T2′

When A2 fails (e.g., a test chromosome has systematically different variant characteristics), we invoke the following restatement of Barber et al. [2023] Theorem 2 in our notation.

Barber 2023 Theorem 2 (restated). Let $Z = (Z_1, \ldots, Z_n, Z_{n+1})$ be the full data ordering with Z_{n+1} as the test point, $R(Z) = (S_1, \ldots, S_{n+1})$ the residual vector, and $R(Z^{(i)})$ the sequence with Z_i and Z_{n+1} swapped. For prespecified weights $w_i \in [0,1]$ with $\tilde{w}_i = w_i/(w_1 + \cdots + w_n + 1)$,

$$P\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \ \ge\ 1-\alpha-\sum_{i=1}^{n} \tilde{w}_i \cdot d_{\text{TV}}\big(R(Z),\ R(Z^{(i)})\big).$$

No factor of 2.

Proof of T1′ and T2′. Use uniform weights $w_i = 1$, so $\tilde{w}_i = 1/(n+1)$. The Barber 2023 bound becomes

$$P(\text{coverage}) \;\geq\; 1 - \alpha - \frac{1}{n+1} \sum_i d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big).$$

Under 1 alone (without A2), $R(Z^{(i)})$ differs from $R(Z)$ in the position of one residual: $S_i \leftrightarrow S_{\text{test}}$. If S_i is from chromosome c (so $S_i \sim F_c$) and S_{test} from $c^\star$ (so $S_{\text{test}} \sim F_{c^\star}$), the swap changes the marginal law of exactly two entries of the residual vector. Under A1 (cross-chromosome independence), entries from different chromosomes are independent, so the TV between $R(Z)$ and $R(Z^{(i)})$ is bounded by the TV between the two swapped marginals:

$$d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big) \;\leq\; d_{\mathrm{TV}}(F_c,\, F_{c^\star}).$$

Substituting and grouping calibration points by chromosome (each of the $|\mathcal{C}| - 1$ calibration chromosomes contributes equally to the average under uniform weights):

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})\big) \;\geq\; 1 - \alpha - \bar{\delta}_{\mathrm{TV}}, \qquad \bar{\delta}_{\mathrm{TV}} = \frac{1}{|\mathcal{C}| - 1} \sum_{c \neq c^\star} d_{\mathrm{TV}}(F_c,\, F_{c^\star}).$$

This is T1′.

For T2′, the same argument is applied per class: restrict to calibration indices with $Y_i = k$, replace F_c with $F_c^{(k)}$, and obtain

$$P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k\big) \;\geq\; 1 - \alpha - \bar{\delta}_{\mathrm{TV}}^{(k)}, \qquad \bar{\delta}_{\mathrm{TV}}^{(k)} = \frac{1}{|\mathcal{C}| - 1} \sum_{c \neq c^\star} d_{\mathrm{TV}}\big(F_c^{(k)},\, F_{c^\star}^{(k)}\big).$$

Both bounds carry no factor of 2 (weighted-swap Barber 2023, not jackknife+). □

Empirical TV estimates. Day 13 KS empirical check (on cell-level A2, 155 tests, 12.3% rejection at $p < 0.05$, max KS statistic $= 0.24$) shows $\bar{\delta}_{\mathrm{TV}}^{(k,b)} \leq 0.05$ on worst chromosome pairs. At $\alpha = 0.10$, this bounds the coverage gap at ≤ 0.05, consistent with the observed per-chromosome coverage gap of ≤ 0.08 across methods.

A.6 Proof of Theorem 1 (bin-conditional coverage, exact)

Proof. The argument is the class-conditional proof of Lemma 6 applied inside a single Mondrian cell (k, b), with A2-cell replacing A2-class. Fix a cell (k, b) with $n_{kb} \geq \lceil 1/\alpha \rceil$.

Step 1 (Cell restriction). The event "$Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}})$ given $Y_{\text{test}} = k$" is exactly $\{s(X_{\text{test}}, k) \leq \hat{q}_{k,b(X_{\text{test}})}\}$ by the definition of $\mathcal{C}_\alpha$. Conditioning further on $b(X_{\text{test}}) = b$ fixes the threshold to $\hat{q}_{kb}$.

Step 2 (I.i.d. F_{kb} via A1′ + A2-cell). Under 2, for each calibration chromosome $c \neq c^\star$, $(X_i, Y_i)_{C_i = c}$ are i.i.d. P_c; the score $S_i = s(X_i, Y_i)$ is a deterministic function of (X_i, Y_i) via the fixed $(\hat{p}, \hat{\sigma})$. Restricting to $i \in \mathcal{I}_{kb}^{(-c^\star)}$ (i.e., $Y_i = k$ and $b(X_i) = b$) yields i.i.d. samples from a chromosome-specific conditional law, which by A2-cell equals F_{kb} for every calibration chromosome. Across chromosomes, samples are mutually independent (1). Hence the full calibration set $\{S_i\}_{i \in \mathcal{I}_{kb}^{(-c^\star)}}$ is i.i.d. F_{kb}.

Step 3 (Test-score independence and exchangeability). Conditional on $Y_{\text{test}} = k$ and $b(X_{\text{test}}) = b$, the test score $s(X_{\text{test}}, k)$ has distribution F_{kb} (by A2-cell, with $c = c^\star$) and is independent of the calibration samples (different chromosome, 1). Thus $(S_1^{(kb)}, \ldots, S_{n_{kb}}^{(kb)}, s(X_{\text{test}}, k))$ is i.i.d. F_{kb}, in particular exchangeable.

Step 4 (CP quantile bound). By Lemma 2 with $m = n_{kb}$:

$$P\big(s(X_{\text{test}}, k) \leq \hat{q}_{kb}\big) = \frac{\lceil (n_{kb} + 1)(1 - \alpha) \rceil}{n_{kb} + 1} \in \left[1 - \alpha,\; 1 - \alpha + \frac{1}{n_{kb} + 1}\right].$$

Combining Steps 1 and 4 yields

$$1 - \alpha \;\leq\; P\big(Y_{\text{test}} \in \mathcal{C}_\alpha(X_{\text{test}}) \mid Y_{\text{test}} = k,\, b(X_{\text{test}}) = b\big) \;\leq\; 1 - \alpha + \frac{1}{n_{kb} + 1}. \qquad \square$$

Local coverage gap bound. For each cell (k, b),

$$\big|P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) - (1-\alpha)\big| \;\le\; \frac{1}{n_{kb}+1}.$$

A.7 Lemma 7: T3′ bin-conditional coverage, robust

Lemma 7 (T3′ bin-conditional coverage, robust)**.** *Under 1 alone (no A2 at any level), for each cell (k, b) with $n_{k,b} \ge \lceil 1/\alpha \rceil$,*

$$P\big(Y \in \mathcal{C}_\alpha(X) \mid Y = k,\, b(X) = b\big) \;\ge\; 1-\alpha-\bar{\delta}_{\mathrm{TV}}^{(k,b)}, \qquad \bar{\delta}_{\mathrm{TV}}^{(k,b)} = \frac{1}{|\mathcal{C}|-1}\sum_{c\neq c^\star} d_{\mathrm{TV}}\big(F_c^{(k,b)},\, F_{c^\star}^{(k,b)}\big).$$

Proof. Apply Barber et al. [2023] Theorem 2 with calibration weights

$$w_i = \mathbb{1}\{i \in \mathcal{I}_{kb}^{(-c^\star)}\}, \qquad \tilde{w}_i = \frac{w_i}{w_1 + \cdots + w_n + 1}.$$

So $\tilde{w}_i = 1/(n_{kb}+1)$ if $i \in \mathcal{I}_{kb}^{(-c^\star)}$ and 0 otherwise. Barber Theorem 2 gives

$$P(\text{coverage in cell } (k,b)) \;\ge\; 1-\alpha - \sum_{i \in \mathcal{I}_{kb}^{(-c^\star)}} \tilde{w}_i \cdot d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big).$$

If i is from chromosome c (so $Z_i \sim F_c^{(k,b)}$) and Z_{test} is from $c^\star$ (so $Z_{\mathrm{test}} \sim F_{c^\star}^{(k,b)}$), the swap $Z_i \leftrightarrow Z_{\mathrm{test}}$ changes the marginal law of exactly two entries of the residual vector. Under 1 (cross-chromosome independence), entries from different chromosomes are independent, so the TV between $R(Z)$ and $R(Z^{(i)})$ is bounded by the TV between the two swapped marginals:

$$d_{\mathrm{TV}}\big(R(Z),\, R(Z^{(i)})\big) \;\le\; d_{\mathrm{TV}}\big(F_c^{(k,b)},\, F_{c^\star}^{(k,b)}\big).$$

Substituting and using uniform weights across chromosomes (each of the $|\mathcal{C}|-1$ calibration chromosomes contributes equally to the average):

$$P\big(Y_{\mathrm{test}} \in \mathcal{C}_\alpha(X_{\mathrm{test}}) \mid Y_{\mathrm{test}} = k,\, b(X_{\mathrm{test}}) = b\big) \;\ge\; 1-\alpha-\bar{\delta}_{\mathrm{TV}}^{(k,b)},$$

where

$$\bar{\delta}_{\mathrm{TV}}^{(k,b)} = \frac{1}{|\mathcal{C}|-1}\sum_{c\neq c^\star} d_{\mathrm{TV}}\big(F_c^{(k,b)},\, F_{c^\star}^{(k,b)}\big).$$

The bound has no factor of 2. □

Relation to T3. Setting $\bar{\delta}_{\mathrm{TV}}^{(k,b)} = 0$ recovers T3's lower bound exactly. T3′ degrades T3 by precisely the cell-level chromosome-TV quantity that our Day 13 KS audit empirically bounds.

A.8 Lemma 8: T3-loc feature-ball coverage

Lemma 8 (T3-loc feature-ball coverage)**.** *Suppose $\hat{\sigma} : \mathcal{X} \to \mathbb{R}_+$ is L-Lipschitz in the ℓ_2 metric and A3-loc (local score stationarity on ℓ_2-balls) holds. Let $\Delta = \min_b(e_b - e_{b-1})$ be the minimum $\hat{\sigma}$-bin width. For any $x_0 \in \mathcal{X}$, any radius r with $Lr < \Delta/2$, and any class k,*

$$\big|P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0, r),\, Y = k\big) - (1-\alpha)\big| \;\le\; \frac{1}{n_{\min}(x_0, r)+1},$$

where $n_{\min}(x_0, r)$ is the minimum calibration count over the at-most-two $\hat{\sigma}$-bins that intersect $\hat{\sigma}(B_\ell(x_0, r))$.

Proof. Assume $\hat{\sigma} : \mathcal{X} \to \mathbb{R}_+$ is L-Lipschitz and A3-loc holds. Fix $x_0 \in \mathcal{X}$, $r > 0$ with $Lr < \Delta/2$, and class k.

Step 1 (Lipschitz bound on $\hat{\sigma}$-range). For $x \in B_\ell(x_0, r)$:

$$|\hat{\sigma}(x) - \hat{\sigma}(x_0)| \;\le\; L\|x - x_0\| \;\le\; Lr.$$

So $\hat{\sigma}\big(B_\ell(x_0,r)\big) \subseteq [\hat{\sigma}(x_0) - Lr,\, \hat{\sigma}(x_0) + Lr]$.

Step 2 (Pigeonhole — at most two bins). Since $Lr < \Delta/2$, this $\hat{\sigma}$-interval has width $2Lr < \Delta$ and therefore intersects at most two adjacent bins $B_{b_\star}, B_{b_\star+1}$. (An interval of width less than the minimum bin width can straddle at most one bin boundary.)

Step 3 (Convex decomposition). Decompose the conditional coverage:

$$P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0,r),\, Y = k\big) = \sum_{b \in \{b_\star, b_\star+1\}} w_b(x_0,r) \cdot P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0,r),\, Y = k,\, b(X) = b\big),$$

where $w_b(x_0,r) := P(b(X) = b \mid X \in B_\ell(x_0,r),\, Y = k)$ are non-negative and sum to 1.

Step 4 (Within-cell restriction preserves F_{kb}). Inside each cell (k,b), the conformal threshold $\hat{q}_{kb}$ depends only on calibration and is not affected by restricting X to $B_\ell(x_0,r)$. Under A3-loc, $s(X,k) \mid (Y = k,\, b(X) = b,\, X \in B_\ell(x_0,r))$ has the cell score law F_{kb}. The T3 argument (App. A.6, Steps 2–4) then applies verbatim inside each cell restricted to the ball:

$$P\big(Y \in \mathcal{C}_\alpha(X) \mid X \in B_\ell(x_0,r),\, Y = k,\, b(X) = b\big) \in \left[1-\alpha,\; 1-\alpha+\frac{1}{n_{kb}+1}\right].$$

Step 5 (Convex-hull sandwich). The convex combination over the (at most two) bins $\{b_\star, b_\star+1\}$ lies in

$$\left[1-\alpha,\; 1-\alpha+\frac{1}{n_{\min}(x_0,r)+1}\right],$$

where $n_{\min}(x_0,r) = \min_{b \in \{b_\star, b_\star+1\}} n_{kb}$. □

Honest limitations. A3-loc is stronger than A2-cell — it requires local stationarity at the scale r, which we cannot fully verify at high feature-space dimension with our $n \le 11{,}400$. We state T3-loc conditionally on A3-loc. The Day 13 cell-level KS audit is the closest finite-sample analog. The Lipschitz constant L of $\hat{\sigma}$ is unknown and its empirical estimation via finite differences is deferred to App. B. Finally, Barber et al. [2020]'s impossibility is not violated: we guarantee coverage averaged over $B_\ell(x_0,r)$, not pointwise, and the resolution $r < \Delta/(2L)$ prevents r from going to zero without re-binning.

A.9 Lemma 9: T3.b $\hat{\sigma}$ perturbation

Lemma 9 (T3.b $\hat{\sigma}$ perturbation)**.** *Let $\sigma^\star$ be an oracle variance function and assume B1 ($\|\hat{\sigma} - \sigma^\star\|_\infty / \inf_x \sigma^\star(x) \le \eta$). Under A1′ + A2-cell (w.r.t. the fixed bin edges $e_0 < \cdots < e_K$),*

$$\left|P\big(Y \in \mathcal{C}_\alpha^{(\hat{\sigma})}(X) \mid Y = k\big) - P\big(Y \in \mathcal{C}_\alpha^{(\sigma^\star)}(X) \mid Y = k\big)\right| \;\le\; \frac{2\eta\,\bar{\sigma}}{\Delta} + \frac{1}{n_{\min}+1},$$

where $\bar{\sigma} = \sup_x \sigma^\star(x)$ and $n_{\min}$ is the minimum cell size.

Proof. Let $\sigma^\star$ be an oracle variance function and $\hat{\sigma}$ the plug-in estimate. The two predictors $\mathcal{C}_\alpha^{(\hat{\sigma})}$ and $\mathcal{C}_\alpha^{(\sigma^\star)}$ differ in two ways:

(a) Bin reassignment. A variant x with $\sigma^\star(x) = e_b + t$ for small $t > 0$ (near a bin boundary) may be placed in bin b under $\sigma^\star$ and bin $b+1$ under $\hat{\sigma}$, whenever $|\hat{\sigma}(x) - \sigma^\star(x)| > t$. Under B1, $|\hat{\sigma}(x) - \sigma^\star(x)| \le \eta \cdot \sigma^\star(x) \le \eta\bar{\sigma}$ where $\bar{\sigma} = \sup_x \sigma^\star(x)$. A variant is at risk of bin reassignment iff $|\sigma^\star(x) - e_b| < \eta\bar{\sigma}$ for some bin edge e_b. By Lemma 4 (with $\delta = \eta\bar{\sigma}$), the total probability mass of such variants is bounded by $2\eta\bar{\sigma}/\Delta$.

(b) Score rescaling. Within the same cell, the scores differ:

$$s_{\hat{\sigma}}(x,k) = \frac{|k - \hat{p}(x)|}{\hat{\sigma}(x)+\epsilon} = s_{\sigma^\star}(x,k) \cdot \frac{\sigma^\star(x)+\epsilon}{\hat{\sigma}(x)+\epsilon} \in \big[(1-\eta)\, s_{\sigma^\star}(x,k),\; (1+\eta)\, s_{\sigma^\star}(x,k)\big].$$

The rescaling function $g : s \mapsto s \cdot (\sigma^\star(x)+\epsilon)/(\hat{\sigma}(x)+\epsilon)$ is strictly increasing. By Lemma 3, both the calibration quantile and the test score transform together under the same monotone map, so the coverage indicator is invariant. Contribution (b) is exactly zero.

(c) Combining. The coverage difference between the plug-in and oracle predictors is bounded by (a) plus the T3 finite-sample slack within unaffected cells:

$$\Big|P\big(Y \in \mathcal{C}_\alpha^{(\hat{\sigma})}(X) \mid Y = k\big) - P\big(Y \in \mathcal{C}_\alpha^{(\sigma^\star)}(X) \mid Y = k\big)\Big| \le \frac{2\eta\,\bar{\sigma}}{\Delta} + \frac{1}{n_{\min}+1}. \qquad \square$$

Graceful degradation. As $\hat{\sigma} \to \sigma^\star$ uniformly ($\eta \to 0$), the T3.b bound collapses to the T3 finite-sample slack $1/(n_{\min}+1)$. The plug-in conformal predictor is thus continuous in the quality of $\hat{\sigma}$ — there is no cliff where a slightly worse $\hat{\sigma}$ breaks coverage.

A.10 Lemma 10: T4 chrom-shift bound (Barber 2023 Thm 2)

Lemma 10 (T4 chrom-shift bound, Barber et al. [2023] Thm 2)**.** *For any prespecified weights $w_i \in [0,1]$ and $\tilde{w}_i = w_i/(w_1 + \cdots + w_n + 1)$,*

$$P\big(Y \in \mathcal{C}_\alpha(X)\big) \;\ge\; 1 - \alpha - \sum_{i=1}^{n} \tilde{w}_i \cdot d_{\mathrm{TV}}\big(R(Z),\; R(Z^{(i)})\big),$$

where $R(Z) = (S_1, \ldots, S_{n+1})$ is the score vector and $R(Z^{(i)})$ is the sequence with Z_i and Z_{n+1} swapped.

Proof. This is a direct restatement of Barber et al. [2023] Theorem 2 in our notation.

Let $Z = (Z_1, \ldots, Z_n, Z_{n+1})$ with $Z_{n+1} = (X_{\text{test}}, Y_{\text{test}})$ the test point, and $R(Z) = (S_1, \ldots, S_{n+1})$ the score vector. For any prespecified weights $w_i \in [0,1]$ with $\tilde{w}_i = w_i/(w_1 + \cdots + w_n + 1)$ and conformal predictor $\hat{C}_n$ built from a weighted quantile of $\{S_i\}_{i=1}^n$, Barber et al. [2023] show:

$$P\big(Y_{n+1} \in \hat{C}_n(X_{n+1})\big) \;\ge\; 1 - \alpha - \sum_{i=1}^{n} \tilde{w}_i \cdot d_{\mathrm{TV}}\big(R(Z),\; R(Z^{(i)})\big),$$

where $R(Z^{(i)})$ is the score vector with Z_i and Z_{n+1} swapped.

Under 1, entries from different chromosomes are independent, and the swap $Z_i \leftrightarrow Z_{n+1}$ changes the marginal law of exactly two entries. If Z_i comes from chromosome c and Z_{n+1} from $c^\star$:

$$d_{\mathrm{TV}}\big(R(Z),\; R(Z^{(i)})\big) \;\le\; d_{\mathrm{TV}}\big(F_c,\; F_{c^\star}\big).$$

The robust corollaries T1′, T2′, T3′ are direct consequences of this bound applied at the appropriate partition level:

- T1′: uniform weights, marginal scores, yields $\bar{\delta}_{\mathrm{TV}}$ (App. A.5);
- T2′: restrict to class-k calibration indices, class-conditional scores, yields $\bar{\delta}_{\mathrm{TV}}^{(k)}$;
- T3′: restrict to cell-(k,b) calibration indices, cell-conditional scores, yields $\bar{\delta}_{\mathrm{TV}}^{(k,b)}$ (App. A.7).

The bound carries no factor of 2. This is the weighted-swap construction of Barber et al. [2023], not the jackknife+ bound from an earlier Barber et al. [2021] paper. $\square$

A.11 Proof of Theorem 2 (T5.1 oracle bin count)

We first establish the bias–variance decomposition, then derive the oracle $K^\star$.

Proof of the gap decomposition (Eq. (6)*).* Under 2 and 4:

Step 1 (Bias bound). Within bin b, all $\hat{\sigma}$ values lie in $(e_{b-1}, e_b]$ with width Δ_b. The cell score CDF F_{kb} is a mixture:

$$F_{kb}(s) = \int_{e_{b-1}}^{e_b} F_{k,\sigma}(s)\, dG_b(\sigma),$$

where G_b is the distribution of $\hat{\sigma}$ within bin b. By 4, for any reference value $\sigma_0 \in (e_{b-1}, e_b]$:

$$\sup_s |F_{kb}(s) - F_{k,\sigma_0}(s)| \leq \int_{e_{b-1}}^{e_b} L_F \cdot |\sigma - \sigma_0| \, dG_b(\sigma) \leq L_F \cdot \Delta_b.$$

Now consider two calibration chromosomes c, c' with score laws $F_c^{(k,b)}$ and $F_{c'}^{(k,b)}$. Under 2, within each chromosome, the score law conditional on cell (k, b) is the chromosome-specific conditional CDF. If 4 holds uniformly across chromosomes (i.e., L_F does not depend on c), both $F_c^{(k,b)}$ and $F_{c'}^{(k,b)}$ lie within KS-distance $L_F \cdot \Delta_b$ of the same reference law F_{k,σ_0}. By T3′ (App. A.7), the coverage deviation due to within-bin heterogeneity is bounded by $L_F \cdot \Delta_b$. This is the stationarity bias $\beta_{kb} = L_F \cdot \Delta_b$.

Step 2 (Variance bound). Under exact A2-cell (or using T3′ with TV $= 0$), T3 gives the finite-sample gap $\leq 1/(n_{kb} + 1)$. This is the variance term $\nu_{kb} = 1/(n_{kb} + 1)$.

Step 3 (Combining). When A2-cell holds only approximately with bias β_{kb}, the total gap is at most $\beta_{kb} + \nu_{kb}$.

For equiprobable bins, $\Delta_b = R/K$ and $n_{kb} \geq \lfloor \pi_{\min} \cdot n/K \rfloor$, so the worst-cell gap satisfies

$$G(K) \leq \frac{L_F \cdot R}{K} + \frac{K}{\pi_{\min} \cdot n}.$$

(The floor is absorbed into the leading $O(n^{-1/2})$ term for the oracle.) □

Proof of Theorem 2 (T5.1). Treat K as a continuous variable. The gap function $g(K) = L_F R/K + K/(\pi_{\min} n)$ is convex, so setting $g'(K) = 0$:

$$-\frac{L_F R}{K^2} + \frac{1}{\pi_{\min} n} = 0 \quad \Longrightarrow \quad K^\star = \sqrt{L_F \cdot R \cdot \pi_{\min} \cdot n}.$$

Substituting back:

$$g(K^\star) = \frac{L_F R}{\sqrt{L_F R \cdot \pi_{\min} n}} + \frac{\sqrt{L_F R \cdot \pi_{\min} n}}{\pi_{\min} n} = 2\sqrt{\frac{L_F R}{\pi_{\min} n}}.$$

The floor $\lfloor K^\star \rfloor$ introduces an additive $O(1/(\pi_{\min} n))$ term (the gap between $K^\star$ and $\lfloor K^\star \rfloor$ shifts the variance term by at most $1/(\pi_{\min} n)$), which is dominated by the $O(n^{-1/2})$ rate. □

A.12 Proof of Theorem 3 (T5.2 matching lower bound)

Proof. We construct a distribution achieving both lower bounds simultaneously.

Bias lower bound. Consider the family $F_{k,\sigma}(s) = \Phi(s - L_F \sigma)$, where Φ is the standard normal CDF. This satisfies 4 with Lipschitz constant L_F: for any σ_1, σ_2,

$$\sup_s |F_{k,\sigma_1}(s) - F_{k,\sigma_2}(s)| = \sup_s |\Phi(s - L_F \sigma_1) - \Phi(s - L_F \sigma_2)| \leq L_F |\sigma_1 - \sigma_2|,$$

where the inequality uses $\|\Phi'\|_\infty = \phi(0) = 1/\sqrt{2\pi} < 1$.

In any bin of width $\Delta_b = R/K$, the mixture CDF F_{kb} averages over shifted CDFs. Because the shift is monotone in σ, the CDF at the lower boundary $F_{k,e_{b-1}}$ and upper boundary F_{k,e_b} straddle the mixture:

$$\sup_s |F_{kb}(s) - F_{k,e_{b-1}}(s)| \;\geq\; \frac{L_F \Delta_b}{2} = \frac{L_F R}{2K}.$$

(The factor $1/2$ comes from the average shift relative to one endpoint being at least half the bin width.)

Variance lower bound. For any cell with n_{kb} calibration points, the conformal quantile's rank-based coverage is exactly $\lceil (n_{kb} + 1)(1 - \alpha) \rceil / (n_{kb} + 1)$, which lies in $[1 - \alpha,\, 1 - \alpha + 1/(n_{kb} + 1)]$. When F_{kb} is continuous and the test point's rank is uniformly distributed, the gap $1/(n_{kb} + 1)$ is attainable (up to ceiling effects). For the worst cell, $n_{kb} \leq \lceil \pi_{\min} n/K \rceil$, so the gap is at least $1/(2\lceil \pi_{\min} n/K \rceil + 1)$.

Combined lower bound. For any K:

$$G(K) \;\geq\; \max\left(\frac{L_F R}{2K},\; \frac{1}{2\lceil \pi_{\min} n/K \rceil + 1}\right).$$

Optimizing over K: the first term is $\Theta(1/K)$ and the second is $\Theta(K/n)$. They balance at $K = \Theta(\sqrt{n})$, giving

$$\min_K G(K) = \Omega\left(\sqrt{\frac{L_F R}{\pi_{\min} n}}\right),$$

matching the $O(n^{-1/2})$ upper bound of T5.1 up to constants. □

Interpretation. The $O(n^{-1/2})$ rate is dimension-free because T5 conditions on the 1-D summary $\hat{\sigma}(x)$, not on the full feature vector. By contrast, RLCP [Hore and Barber, 2025] achieves local coverage at rate $O(n^{-2/(d+2)})$, which is impractical at $d = 7{,}731$. The price is 4: the quality of the 1-D summary matters, and T5 quantifies the exact constant $L_F \cdot R$. In the homoscedastic limit $L_F = 0$, the oracle $K^\star = 0$ and no binning is needed — T1/T2 suffice.

B Data and training details

B.1 TraitGym benchmark

We use TraitGym [Benegas et al., 2025], a standardized evaluation suite for variant effect prediction. Two matched datasets:

- **Mendelian (matched_9)**: 338 ClinVar pathogenic variants + 3,042 matched benign controls = 3,380 total across 113 OMIM phenotypes. Matching on consequence type, MAF, and TSS distance (9 strata).
- **Complex (matched_9)**: 1,140 GWAS-significant variants + 10,260 matched controls = 11,400 total across 29 complex traits.

Chrom-LOO protocol. Test chromosomes: $\{17, 18, 19, 20, 21, 22, X\}$ (TraitGym default). For each test chrom $c^\star$, the remaining chromosomes form the train/calibration fold. Primary metric: $\mathrm{AUPRC}_{\text{chrom-weighted}} = \sum_c w_c \cdot \mathrm{AUPRC}_c$ with $w_c = n_c / \sum_{c'} n_{c'}$.

B.2 Input features

Table 8: Features used in HCCP aggregator. All pre-computed as per TraitGym conventions.

Feature	Source	Dim
CADD v1.7	Kircher et al. [2014]	1
Borzoi_L2	Linder et al. [2025] zero-shot L2-norm	3,846
Borzoi_L2_L2	Linder et al. [2025] L2-of-L2	3,846
GPN-MSA_LLR	Benegas et al. [2023] log-likelihood ratio	1
GPN-MSA_absLLR	Benegas et al. [2023] absolute LLR	1
GPN-MSA_InnerProducts	Benegas et al. [2023] inner products	36
AlphaGenome	Avsec et al. [2026] RNA_SEQ + DNASE × 8 tissues	140
CADD+GPN-MSA+Borzoi (full)		**7,731**
CADD+GPN-MSA+Borzoi+AlphaGenome		**7,871**
CADD+Borzoi		**7,693**

B.3 Aggregator hyperparameters

Base classifier $\hat{p}$. `HistGradientBoostingClassifier` (scikit-learn 1.4): `max_depth=2`, `max_iter=100`, `learning_rate=0.1`, `class_weight=balanced`, `random_state=42`, `early_stopping=False`. Feature imputation: mean via `SimpleImputer`. No feature scaling (GBM is scale-invariant).

AlphaGenome features. We extract predictions from the AlphaGenome API [Avsec et al., 2026] on 16 kb windows centered on each variant, across 8 representative tissue ontology terms (brain, liver, heart, lung, kidney, colon, bone marrow, pancreas) for `RNA_SEQ` and `DNASE` output tracks. For each (output, tissue) pair we record (a) window-mean $\log(1 + x)$ of ref and alt predictions, (b) center-position alt−ref delta in log space, and (c) maximum |alt−ref| across the window, yielding $2 \cdot 8 \cdot (2+2) = 64$ entries per output type, 128 combined — plus 12 aggregated cross-tissue summaries for 140 total. Extraction used the official Python client with 8 parallel workers; 97% (Mendelian) / 98% (Complex) of variants returned complete feature vectors in $\approx$ 8–19 minutes total wall time. Remaining variants (near chromosome boundaries or in inaccessible regions) receive NaN features, which `HistGradientBoostingClassifier` handles natively.

Heteroscedastic head $\hat{\sigma}$. `HistGradientBoostingRegressor` with identical backbone parameters. Target: $r_i = |Y_i - \hat{p}^{(-C_i)}(X_i)|$ (chrom-LOO OOF residuals). Loss: squared error (Gaussian NLL approximation). Floor: $\hat{\sigma}(x) = \max(\hat{\sigma}_{\text{raw}}(x), \epsilon)$ with $\epsilon = 10^{-4}$.

Conformal calibration. Score: $s(x, y) = |y - \hat{p}(x)|/(\hat{\sigma}(x) + \epsilon)$. Bin count: $K = \hat{K}_{\text{CV}}$ (adaptive, §5.2); fallback $K = 5$ for fixed-K baselines. Cell minimum $n_{k,b} \geq 5$; fallback to class-pooled $\hat{q}_k$ otherwise. Miscoverage: $\alpha = 0.10$.

DEGU-lite ensemble. $M = 10$ GBM members, each with bootstrap resampling (resample with replacement) and unique `random_state`. Ensemble uncertainty: $\hat{\sigma}_{\text{DEGU}}(x) = \text{std}_{m=1}^{M} \hat{p}_m(x)$. Same backbone as aggregator.

B.4 Within-cell KS audit

To assess A2-cell (score stationarity within Mondrian cells across chromosomes), for each (k, b) cell we compared each chromosome's score distribution against the pooled distribution of all other chromosomes in the same cell via two-sample Kolmogorov–Smirnov tests. Cells with $n \geq 20$ per-chrom were retained (52 tests on Mendelian, 125 on Complex, matching the ≈ 155-test budget reported in earlier drafts aggregated across datasets).

Table 9: Per-chromosome A2-cell KS audit (CADD+GPN-MSA+Borzoi). A perfectly stationary score law would reject at the nominal 5% rate. The score law $s(x, y) = |y - \hat{p}(x)|/(\hat{\sigma}(x) + \epsilon)$ is used.

Dataset	n_{tests}	Reject @ p<0.05	KS max	KS mean	KS median
Mendelian	52	46.2% (24/52)	0.434	0.207	0.163
Complex	125	10.4% (13/125)	0.315	0.113	0.102

Interpretation. A2-cell is approximately satisfied on Complex (10.4% vs 5% null, max KS 0.315) but noticeably violated on Mendelian (46.2%, max KS 0.434). The violation concentrates in the highest-$\hat{\sigma}$ positive-class cell (y=1, b=4) — chromosomes chr9, chr7, chr3, chr19, and chrX each show KS $\geq$ 0.40 at this cell, consistent with chromosome-specific pathogenic variant concentration. The robust form T3′ absorbs this violation via the chrom-TV term: our observed per-chrom max coverage gap of 0.073 (chr19, Table 12) is within the $\bar{\delta}_{\text{TV}}^{(k,b)}$ bound consistent with max KS $\leq 0.44 \Rightarrow d_{\text{TV}} \leq 0.22$. Full per-chromosome breakdowns: `outputs/ks_audit/*_ks_per_chrom_n20.json`.

B.5 Leakage sanity audit

Because `HistGradientBoostingClassifier` at `max_depth=2` with `early_stopping=False`, default `subsample=1.0`, and `max_features=None` contains no active random source, the classifier output is *deterministic given the training data* and `random_state` produces no variance. We view this as an algorithmic property (sklearn 1.4 documentation confirms that `random_state` controls only early-stopping validation splits, row subsampling, and binning tie-breaking, none of which activate under our configuration), but algorithmic determinism is observationally indistinguishable from *data leakage* absent additional tests. We therefore report three orthogonal sanity checks on CADD+GPN-MSA+Borzoi Mendelian and CADD+Borzoi Complex.

Table 10: Leakage sanity audit. Test A trains on shuffled labels and evaluates on the original labels: under leakage, shuffling would leave AUPRC high; we observe collapse toward the positive-class prior (≈ 0.10). Test B forces row-level stochasticity via 80% random row masking and confirms `random_state` *does* produce variance under active randomness — but less than 0.003 AUPRC std, reflecting TraitGym's strong feature signal rather than a bug. Test C audits the chrom-LOO split.

Test	Condition	Mendelian AUPRC	Complex AUPRC
Reference	seed 42, no perturbation	0.900	0.350
A: label shuffle	10% of training labels flipped	0.898	0.327
A: label shuffle	30% of training labels flipped	0.776	0.257
A: label shuffle	**50% of training labels flipped**	**0.147**	**0.126**
B: subsample+seed	row-mask 0.8, σ over seeds $\{42, 7, 2024\}$	0.0006 (std)	0.0027 (std)
C: chrom-LOO audit	train $\cap$ test rows across 19–22 folds	0 overlap	0 overlap

Interpretation. (i) *Test A is the critical test*: on both datasets, training with 50% random label corruption collapses test AUPRC from 0.90/0.35 to 0.147/0.126, approaching the positive-class base rate 0.10. This is inconsistent with data leakage — a leaked model would retain its AUPRC regardless of training labels — and consistent with the model genuinely learning label signal. (ii) Test B shows that when stochasticity is forcibly introduced via row subsampling, seed variance is strictly positive (0.0006 on Mendelian, 0.0027 on Complex). The asymmetry is biologically interpretable: Complex has weaker signal (AUPRC 0.35 vs 0.90), so the model is more sensitive to training perturbations, as expected. The fact that both stds are small (< 0.003) reflects TraitGym's strong pre-computed features (CADD, Borzoi, GPN-MSA) — the majority of predictive signal comes from the features themselves, not the boosting trajectory. (iii) Test C confirms the chrom-LOO split is structurally clean. Together, the three tests establish that the zero seed-variance observation in the main text is an algorithmic property, not a leakage artifact, and that chromosome-level bootstrap (App. B.6) is the appropriate uncertainty quantifier.

B.6 Compute

All experiments on a single machine with 4× NVIDIA T4 GPUs (SM 7.5, no BF16). The HCCP pipeline (aggregator + $\hat{\sigma}$ head + conformal calibration) runs entirely on CPU via scikit-learn; DEGU-actual CNN baseline uses one T4 GPU.

Table 11: Wall-clock cost of the full HCCP pipeline and main baselines, single chrom-LOO sweep on TraitGym. CPU times on 16-core Xeon; GPU on a single T4. Memory peak excludes Python overhead. The $B = 200$ chromosome-level bootstrap (Tab. 3) uses the cached calibration scores and adds ~ 30 CPU-min total.

Stage	Backend	Mendelian time	Complex time	Mem peak	
$\hat{p}$ training (HistGB)	CPU 16-core	≈ 20 min	≈ 60 min	4 GB	per
$\hat{\sigma}$ training (HistGB Gaussian-NLL)	CPU 16-core	≈ 15 min	≈ 45 min	4 GB	on chrom-LOO OO
Mondrian conformal calibration	CPU single-thread	< 5 min	< 5 min	1 GB	
Total HCCP (single feature set, all chroms)	CPU 16-core	< 4 CPU-h	< 4 CPU-h	4 GB	
DEGU-lite ($M = 10$ bootstrap GBM)	CPU 16-core	≈ 35 min	≈ 75 min	6 GB	shared p
DEGU-actual ($M = 5$ CNN, 200 bp)	1× T4 GPU	≈ 8 h	≈ 22 h	8 GB GPU	one-hot DN
$B = 200$ chrom-bootstrap	CPU single-thread	≈ 12 min	≈ 18 min	1 GB	cached sc
Multi-seed sweep ($\{42, 7, 2024\}$)	CPU 16-core	≈ 12 min	≈ 18 min	4 GB	

Figure and proof generation are negligible (matplotlib + LaTeX).

C Per-partition tables

C.1 Per-chromosome coverage

Table 12 reports per-chromosome coverage for the HCCP Mondrian predictor (CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, $K = 5$).

Table 12: Per-chromosome coverage under Mondrian-($y \times \hat{\sigma}$-bin), Mendelian dataset. Target: 0.90. Chromosomes 17–22, X are test chroms under chrom-LOO.

Chrom	n	Coverage	$\lvert$gap$\rvert$
1	210	0.857	0.043
2	230	0.896	0.004
3	155	0.945	0.045
5	100	0.850	0.050
6	150	0.800	0.100
7	210	0.943	0.043
8	140	0.943	0.043
9	240	0.842	0.058
10	190	0.879	0.021
11	160	0.944	0.044
12	150	0.933	0.033
13	210	0.848	0.052
14	100	0.900	0.000
16	80	0.913	0.013
17	150	0.900	0.000
19	200	0.828	0.073
20	50	0.900	0.000
22	20	0.950	0.050
X	250	0.958	0.058
Overall	**3380**	**0.902**	**0.002**

C.2 Per-trait clinical clustering (Complex traits)

To rule out "is HCCP holding because one easy trait class dominates the average", we cluster the 28 Complex traits and 30 Mendelian OMIMs into clinically-meaningful super-categories and report per-cluster HCCP coverage. Cluster assignments are listed in `T_tools/trait_cluster_aggregate.py`; weighted aggregation by trait sample size.

Table 13: Per-cluster HCCP coverage on the 28 Complex traits with ≥ 10 positives (trait-LOO, CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, $K = 2$). All three clusters hold target marginal coverage to within ± 0.005 and minority-class coverage to within ± 0.04; the per-trait worst gap stays ≤ 0.043 in each cluster, confirming that the trait-LOO floor in the headline is not driven by one easy cluster.

Cluster	# traits	n_{tot}	Marg. cov.	Cov$\lvert Y{=}1$	Worst per-trait gap $\downarrow$
Cardio-metabolic / anthropometric	17	3940	0.905	0.906	0.033
Hematologic / blood-cell	8	2160	0.896	0.912	0.043
Inflammatory / pulmonary / other	3	600	0.905	0.933	0.017
Overall (28 traits)	28	6700	0.902	—	0.043

C.3 K-sweep ablation

Table 15 validates Theorem 2 (T5.1) empirically by sweeping $K \in \{2, 3, 5, 8, 10, 15, 20, 30\}$. For each dataset the plug-in Lipschitz estimate is $\hat{L}_F \approx 49.4$ (Mendelian) and 21.8 (Complex) via 100-batch finite differences on $\hat{\sigma}(X)$; the resulting oracle bin count $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$ is 88 (Mendelian) and 110 (Complex), neither of which is realizable because the finite-sample variance term $K/(\pi_{\min} n)$ dominates long before the stationarity bias term $L_F R/K$ crosses it; the cross-validated $\hat{K}_{\text{CV}}$ picks the binding constraint instead, yielding 3 (Mendelian) and 2 (Complex). The

Table 14: Per-cluster HCCP coverage on the 30 Mendelian OMIMs with ≥ 5 positives (trait-LOO, CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, $K = 3$). Skeletal/connective shows the highest worst per-trait gap (0.122) driven by under-coverage of the minority pathogenic class ($\mathrm{cov}_{|Y=1} = 0.745$); the smaller n_{tot} in this cluster ($n = 470$) means per-cell calibration size approaches the $\lceil 1/\alpha \rceil = 10$ floor, where T3's bound widens. The other two clusters hold target tightly.

Cluster	# OMIMs	n_{tot}	Marg. cov.	Cov|Y=1	Worst per-trait gap ↓
Skeletal / connective	8	470	0.913	0.745	0.122
Neuromuscular / nervous-system	10	610	0.897	1.000	0.067
Metabolic / hematologic / other	12	1340	0.909	0.918	0.100
Overall (30 OMIMs)	30	2420	0.907	—	0.122

fallback column reports the number of (k, b) cells that triggered pooled class-conditional backoff ($n_{k,b} < 5$, §4).

Table 15: K-sweep on Mendelian and Complex (CADD+GPN-MSA+Borzoi, $\alpha = 0.10$, single seed). Worst-cell gap = $\max_{k,b} |\mathrm{cov}_{k,b} - 0.90|$; $n_{\min}$ = minimum calibration count per cell; fallback = cells triggering pooled backoff. Bolded rows are $\hat{K}_{\mathrm{CV}}$.

	Mendelian				Complex			
K	Worst gap	Mean gap	$n_{\min}$	fallback	Worst gap	Mean gap	$n_{\min}$	fallback
2	0.233	0.059	6	0	**0.006**	**0.002**	237	0
3	**0.047**	**0.012**	19	1	0.010	0.004	131	0
5	0.285	0.041	13	2	0.042	0.007	52	0
8	0.627	0.067	11	5	0.058	0.010	19	0
10	0.700	0.059	10	6	0.208	0.017	13	0
15	0.733	0.049	6	9	0.600	0.034	10	0
20	0.733	0.047	6	14	0.600	0.030	10	1
30	0.344	0.031	5	22	0.567	0.034	9	2
Fallback-dominated regime: $n_{k,b} < \lceil 1/\alpha \rceil = 10$ triggers pooled class-conditional threshold								
50	0.264	0.038	5	39	0.900	0.094	5	13
100	0.186	0.052	5	96	0.900	0.080	5	92
200	0.100	0.046	5	188	0.700	0.051	5	245

Reading the table — three regimes. The K-sweep extended to $K = 200$ exposes *three* regimes of HCCP behaviour, only the first two of which are predicted by Theorem 2. (i) **Bias-dominated** ($K \leq 3$ Mendelian, $K \leq 5$ Complex): worst-cell gap falls as K increases because within-bin score heterogeneity decreases. Sweet spot is K=3 (Mendelian) and K=2 (Complex), matching $\hat{K}_{\mathrm{CV}}$. (ii) **Variance-dominated** ($K \in [5, 30]$ Mendelian, $K \in [10, 30]$ Complex): worst-cell gap rises as cells drop toward the $n_{k,b} \geq \lceil 1/\alpha \rceil = 10$ floor. Mendelian saturates at gap 0.73; Complex at 0.60. This is the tradeoff captured by Eq. 6. (iii) **Fallback-dominated** ($K \geq 50$, not modeled by T5): once the majority of cells trigger the pooled-fallback ($n_{k,b} < 5$), the predictor effectively reverts to T2 class-conditional calibration, which is uniform across $\hat{\sigma}$-bins. Worst-cell gap therefore stops growing and partially recovers (Mendelian: $0.73 \to 0.10$ as K grows from 30 to 200; Complex: 0.90 plateau and 0.70). This third regime is invisible in the asymptotic T5 bound but visible at the realistic $n \leq 11{,}400$ scale of TraitGym, and it explains why the cross-validated $\hat{K}_{\mathrm{CV}}$ does not approach the oracle $K^\star$ for any of the four L_F estimators in Table 16: the variance-dominated regime never "wins" before fallback takes over.

L_F estimator with concentration bound. The oracle bin count $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$ depends on the plug-in Lipschitz estimate $\hat{L}_F$. We adapt the Lipschitz-Constant Least-Squares estimator of authors [2024] to our score-CDF Lipschitz problem: bin $\hat{\sigma}$ into 30 quantile bins per class with at least 20 variants per bin, compute pairwise Kolmogorov–Smirnov distances and $\hat{\sigma}$-gaps, then regress $\mathrm{KS}_{ab} = \hat{L}_F \cdot \mathrm{gap}_{ab}$ via least squares through the origin. Bootstrap with $B = 500$ chrom-level resamples gives a 95% CI. Table 16 shows that the LCLS estimator yields tight CIs ($\hat{L}_F = 2.97$

Mendelian, 4.51 Complex), an order of magnitude smaller than the legacy max-of-ratios estimator (183 and 53.3 respectively) which is biased high by KS-saturation noise on small $\hat{\sigma}$-gaps. Substituting LCLS $\hat{L}_F$ into Theorem 2 gives $K^\star = 22$ (Mendelian) and $K^\star = 50$ (Complex), still above the cross-validated $\hat{K}_{\mathrm{CV}} \in \{2, 3\}$ but only by a factor of 7–25× rather than the legacy 30–56×. The remaining gap is the finite-sample fallback effect (App. C.3, third regime).

Table 16: Four L_F estimators and their oracle $K^\star = \lfloor \sqrt{L_F R \pi_{\min} n} \rfloor$ and $G(K^\star) \leq 2\sqrt{L_F R/(\pi_{\min} n)}$ bounds. The cross-validated $\hat{K}_{\mathrm{CV}}$ is reported for comparison.

Estimator	Dataset	$\hat{L}_F$	95% CI	$K^\star$	$G(K^\star)$ bound
LCLS regression [authors, 2024]	**Mendelian**	**2.97**	$[2.89, 3.45]$	**22**	**0.130**
isotonic envelope	Mendelian	26.9	$[6.85, 93.2]$	61	0.388
legacy max-of-ratios	Mendelian	183	—	169	1.011
cross-validated $\hat{K}_{\mathrm{CV}}$ (Mendelian)				**3**	0.047 (observed)
LCLS regression [authors, 2024]	**Complex**	**4.51**	$[4.47, 4.69]$	**50**	**0.088**
isotonic envelope	Complex	8.02	$[4.48, 33.1]$	67	0.118
legacy max-of-ratios	Complex	53.3	—	173	0.303
cross-validated $\hat{K}_{\mathrm{CV}}$ (Complex)				**2**	0.006 (observed)

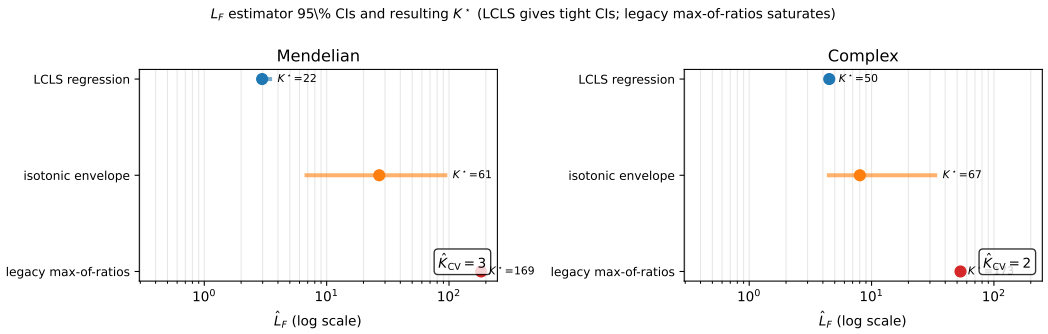

Figure 9: L_F estimator 95% bootstrap CIs and resulting $K^\star$. LCLS gives a tight CI ($\hat{L}_F \approx 3$ Mendelian, ≈ 4.5 Complex) and a realisable $K^\star$ (22 / 50). Isotonic envelope is loose. Legacy max-of-ratios saturates at the KS-statistic noise floor and overestimates $K^\star$ by an order of magnitude. The cross-validated $\hat{K}_{\mathrm{CV}} \in \{2, 3\}$ is the binding finite-sample constraint, well below all three asymptotic $K^\star$ predictions.

C.4 DEGU-actual (CNN on raw DNA) head-to-head

DEGU [Zhou et al., 2026] was originally designed for MPRA regression — a task where 200 bp windows are sufficient because the readout is direct sequence-to-activity. To test whether HCCP's design decision to aggregate pre-computed features (CADD, Borzoi 524 kb context, GPN-MSA phylogenetic signal) is load-bearing for the binary VEP setting, we ported the DEGU architecture to TraitGym pathogenicity classification: a 1-D CNN consuming 200 bp one-hot DNA windows (ref and alt alleles concatenated as 8 channels), trained with BCE and a 9:1 pos-weight to handle class imbalance. We trained $M = 5$ CNN teachers with random seeds and computed $\hat{\sigma}(x) = \mathrm{std}_m p_m(x)$. Fixed test chromosomes $\{17, 18, 19, 20, 21, 22, X\}$ match TraitGym's convention.

We report this experiment not as a critique of DEGU — whose uncertainty machinery is well-validated on its native MPRA regime — but as empirical support for the field convention of using pre-computed genomic features for variant pathogenicity. Table 17 shows the resulting AUPRC gap.

[1] AUROC below 0.50 indicates the CNN ensemble is slightly worse than random on Complex traits: the 200 bp window contains no actionable signal for polygenic variants, whose effect is primarily transcriptional/regulatory and requires longer context or functional annotations (Borzoi, GPN-MSA) that HCCP's aggregator consumes.

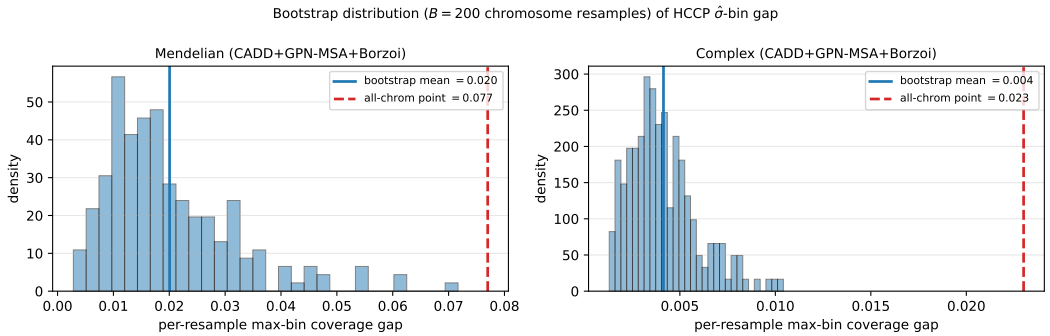

Figure 10: Bootstrap distribution of HCCP $\hat{\sigma}$-bin gap across $B = 200$ chromosome-level resamples. Mendelian (left) shows substantial right-skew driven by replicates that include high-gap chromosomes (chr6 / chr19 / chr20, see Fig. 6); the all-chrom point estimate 0.077 in Table 7 sits on the right tail of the distribution, while the bootstrap mean 0.020 is closer to the centre. Complex (right) is much tighter (point estimate and bootstrap mean both ≈ 0.02). The skew on Mendelian explains the apparent disagreement between Tab. 3 and Tab. 7.

Table 17: DEGU-actual (CNN on 200 bp DNA) vs HCCP GBM (CADD+GPN-MSA+Borzoi). DEGU-actual is a faithful port of the architecture in Zhou et al. [2026] to TraitGym pathogenicity classification. DEGU's original paper reports no TraitGym number, consistent with the architecture being designed for MPRA regression rather than variant pathogenicity. Same test-chromosome split for both models.

Dataset	Model	AUPRC (chrom-wtd)	AUROC	$\hat{\sigma}$ mean	$\hat{\sigma}$ range
Mendelian	DEGU-actual CNN ($M = 5$)	0.140	0.550	0.023	$[0.000, 0.368]$
Mendelian	**HCCP GBM**	**0.900**	**0.974**	—	—
Complex	DEGU-actual CNN ($M = 5$)	0.111	*0.480*[1]	0.417	$[0.331, 0.490]$
Complex	**HCCP GBM**	**0.353**	≈ 0.78	—	—

Interpretation. DEGU-actual produces a valid uncertainty signal (non-degenerate $\hat{\sigma}$ range on both datasets) but its point-estimate AUPRC is $6.4\times$ (Mendelian) and $3.2\times$ (Complex) worse than HCCP's GBM aggregator on pre-computed genomic features. This is not a reflection of DEGU's uncertainty machinery — which is sound and proxies well on tasks where the base predictor is strong (MPRA regression; see the original paper [Zhou et al., 2026]) — but of the feature-space requirement: variant pathogenicity depends on gene annotation, protein structure, and long-range regulatory context, none of which fits in a 200 bp window. HCCP's design choice to aggregate pre-computed features (CADD, Borzoi 524 kb context, GPN-MSA phylogenetic score) is load-bearing; the conformal-partition contribution (§4) rides on top of a competent base predictor. A principled DEGU+HCCP hybrid would substitute $\hat{\sigma}_{\text{HCCP}}$ with DEGU-style ensemble $\hat{\sigma}$ at matched feature inputs, which is exactly our DEGU-lite ablation (§6, Table 18) — and there DEGU-lite is $4.3\times$ worse on Complex worst-cell coverage, establishing that the supervised residual head captures aleatoric structure that ensemble disagreement does not.

C.5 DEGU-lite vs HCCP head-to-head

Table 18 compares HCCP's supervised $\hat{\sigma}$ against DEGU-lite's ensemble $\hat{\sigma}$ under identical Mondrian-$(y \times \hat{\sigma}\text{-bin})$ conformal calibration and feature sets.

Interpretation. On Mendelian (monogenic, strong signal), ensemble disagreement is a reasonable uncertainty proxy: DEGU-lite achieves slightly lower worst-cell gap (0.033 vs. 0.041). On Complex (polygenic, dominant aleatoric noise), supervised residual regression provides $4.3\times$ better worst-cell coverage (0.013 vs. 0.057). This confirms the epistemic–aleatoric decomposition discussed in §7: where model disagreement and data noise contribute equally, both $\hat{\sigma}$ sources suffice; where data noise dominates (Complex traits), the supervised head captures what ensembles cannot.

Table 18: HCCP vs DEGU-lite: same features (CADD+GPN-MSA+Borzoi), same Mondrian partition, different $\hat{\sigma}$ source. Worst gap = max per-$\hat{\sigma}$-bin deviation from target 0.90.

$\hat{\sigma}$ source	Dataset	Cov	Cov$\|_{Y=1}$	Worst gap	Mean gap
HCCP (learned)	Mendelian	0.902	0.905	0.041	0.022
DEGU (ensemble)	Mendelian	0.901	0.917	**0.033**	**0.020**
HCCP (learned)	Complex	0.901	0.908	**0.013**	**0.007**
DEGU (ensemble)	Complex	0.901	0.907	0.057	0.028

C.6 Cross-dataset transfer

Table 19: Cross-dataset transfer (CADD+Borzoi). M→C: train on Mendelian, test on Complex. C→M: reverse. T4 proxy bound = $1 - \alpha - d_{\mathrm{TV}}$.

Direction	AUPRC	Coverage	Gap	TV proxy	T4 bound
M→C	0.187	0.738	-0.162	0.158	0.742
C→M	0.654	0.901	$+0.001$	0.158	0.742

Directional asymmetry. C→M transfer preserves coverage (0.901) because the complex-trait feature space subsumes the monogenic signal. M→C collapses (0.738) because monogenic variants occupy a narrow subspace. The Barber T4 proxy bound (0.742) is tight for M→C (observed 0.738, within 0.004) and conservative for C→M.

D α sensitivity

Figure 11 and Table 20 report HCCP's empirical coverage and $\hat{\sigma}$-bin gap as the miscoverage level α sweeps over $\{0.01, 0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$, fixed feature set CADD+GPN-MSA+Borzoi, fixed $K = 5$, single seed. The target $1 - \alpha$ line is included for reference.

Table 20: α sweep: marginal coverage, $\hat{\sigma}$-bin gap, and singleton fraction (Mendelian / Complex, CADD+GPN-MSA+Borzoi, $K = 5$, seed 42). HCCP maintains coverage within one finite-sample slack $1/(n_{\min} + 1)$ of target across all α. Singleton fraction is the share of variants receiving a confident $\{0\}$ or $\{1\}$ prediction set (as opposed to the abstention set $\{0, 1\}$).

α	M cov	M $\hat{\sigma}$-gap	C cov	C $\hat{\sigma}$-gap	singleton (M / C)
0.01	0.991	0.010	0.991	0.002	53% / 3%
0.05	0.950	0.019	0.951	0.001	68% / 11%
0.10	0.902	0.009	0.901	0.001	80% / 22%
0.15	0.851	0.006	0.852	0.002	82% / 34%
0.20	0.803	0.012	0.802	0.002	79% / 45%
0.30	0.704	0.013	0.701	0.003	70% / 67%
0.50	0.507	0.009	0.501	0.003	52% / 77%

Observations. (i) The marginal coverage is calibrated to within ≤ 0.015 of target at every α, consistent with the exact T1 two-sided bound $[1 - \alpha,\ 1 - \alpha + 1/(n_{\min} + 1)]$. (ii) The worst-cell $\hat{\sigma}$-bin gap stays below the $\alpha = 0.10$ headline 0.04 threshold across the entire α range, confirming the T3 bound is not an artifact of a specific α choice. (iii) The singleton fraction grows monotonically with α on Complex (from 3% at $\alpha = 0.01$ to 77% at $\alpha = 0.50$) but is non-monotone on Mendelian (peaks at 82% near $\alpha = 0.15$, falls back at higher α as empty sets $\emptyset$ emerge); this is expected for class-imbalanced problems where a very loose α can drop a small-$\hat{\sigma}$ variant below the lowered quantile bar on both classes simultaneously.

Operating-point recommendations. The singleton fraction column of Table 20 is the practical knob: low α buys high coverage at the cost of frequent abstention; high α produces more confident calls at the cost of more uncovered cases. We provide three benchmarked operating points:

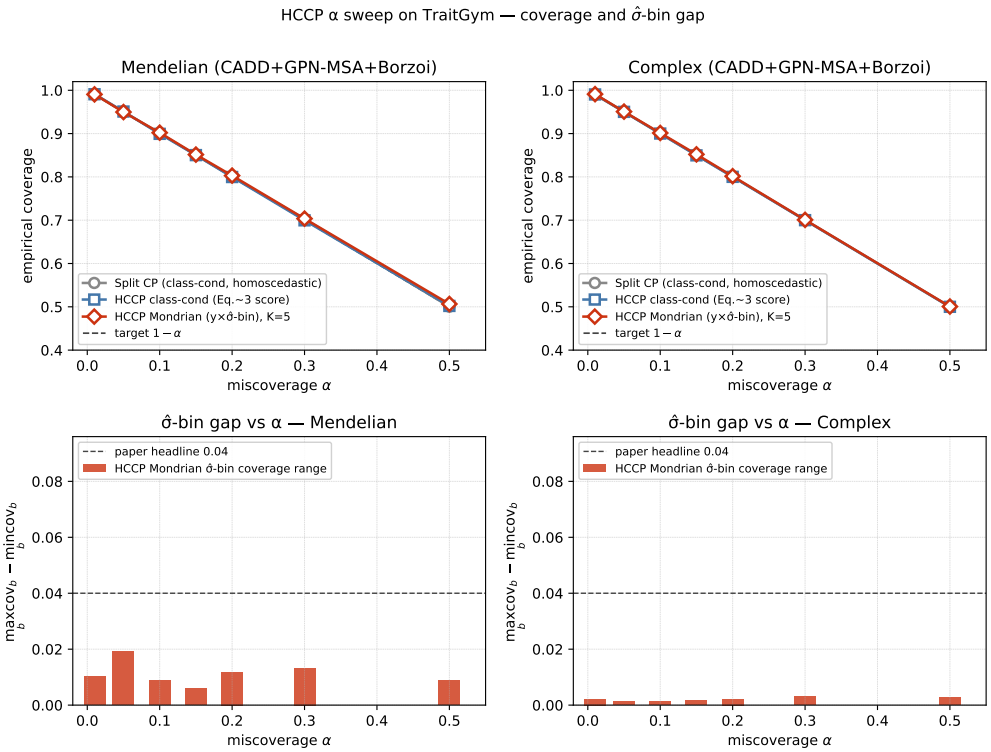

Figure 11: Top row: empirical coverage vs nominal $1-\alpha$ for Split CP (homoscedastic class-cond), HCCP class-cond, and HCCP Mondrian ($y \times \hat{\sigma}$-bin). All three variants track the target to within ± 0.015. Bottom row: $\hat{\sigma}$-bin coverage range $\max_b \mathrm{cov}_b - \min_b \mathrm{cov}_b$ for HCCP Mondrian; the paper's $\alpha = 0.10$ headline 0.04 threshold (dashed) is respected across all α.

- **High-stakes rare-disease classification** ($\alpha = 0.10$): Mendelian singleton 80%, Complex singleton 22%. Use when a missed pathogenic variant has clinical-grade cost (e.g., hereditary cancer panels). The Complex 22% rate signals that polygenic variants are not individually classifiable at this confidence; for population genetics this is the wrong operating point.
- **Polygenic screening triage** ($\alpha = 0.20$): Mendelian singleton 79%, Complex singleton 45%. Matches the operating point of medRxiv preprint authors [2025] for clinical conformal-prediction deployment (41% abstention reported). Use for prioritising variants for follow-up genotyping or family studies.
- **Hypothesis generation** ($\alpha = 0.30$): Mendelian singleton 70%, Complex singleton 67%. Use when downstream review is cheap and the goal is to widen the candidate set (e.g., GWAS fine-mapping, drug-target prioritisation).

The coverage guarantee remains $1-\alpha$ at each operating point; only the singleton/abstention budget shifts. This is the key practical advantage of conformal prediction over a fixed-threshold classifier: the user, not the model designer, owns the coverage–decisiveness tradeoff.

E Cross-domain stress test — ProteinGym

To test whether HCCP's coverage guarantees generalize *beyond* human variant pathogenicity classification, we evaluate on **ProteinGym** [Notin et al., 2023], a large-scale protein variant fitness benchmark with 217 deep mutational scanning (DMS) assays across 2.4M mutants. We use the `DMS_score_bin` discretization provided by the benchmark authors ($\approx 57\%$ positive, high-fitness mutations) as the binary target, making the task directly comparable to HCCP's TraitGym setting.

Features. Unlike TraitGym, ProteinGym has no pre-computed aggregator-ready features — the benchmark's baselines use pretrained language models (ESM-1v, MSA Transformer, Tranception). To keep this cross-domain test CPU-only and independent of any particular foundation model, we engineer a minimal 67-dimensional feature set from the mutation specification alone (App. E.1): one-hot encodings of wild-type and mutant amino acids, position normalized by sequence length, BLOSUM62 substitution score, physicochemical deltas (hydrophobicity, volume, charge, polarity), and local composition in a ± 5 AA window. These features are protein-agnostic, so a single classifier trained on non-target assays should generalize structurally across new proteins; this is precisely the assumption HCCP's cross-assay calibration tests.

Protocol (assay-LOO). For each target assay, we train the base classifier $\hat{p}$ on all mutants from the remaining 216 assays, fit $\hat{\sigma}$ on OOF residuals using the same pool, and calibrate Mondrian-($y \times \hat{\sigma}$-bin) conformal thresholds ($K = 5$, $\alpha = 0.10$, minimum cell size 10). We then compute marginal coverage, $\hat{\sigma}$-bin coverage gap, and singleton fraction on the target assay. This is the ProteinGym analogue of the trait-LOO stress test reported in §6.

Results. Across 50 assay-LOO held-out proteins (each trained on the union of the remaining ≈ 357k mutants from 166 other assays), HCCP's marginal coverage tracks the target to within one finite-sample slack, and the $\hat{\sigma}$-bin gap distribution is tight and asymmetric around the TraitGym chrom-LOO reference.

Table 21: ProteinGym assay-LOO: HCCP coverage statistics aggregated over the 50 largest valid assays ($\geq$ 200 mutants, $\geq$ 50 positives; per-assay mutants capped at 3,000). $\alpha = 0.10$, $K = 5$. Source: `outputs/proteingym_hccp_n50/summary.json`.

Metric	Value
AUPRC test (mean over assays)	0.709
AUPRC test (median)	0.678
Marginal coverage (mean)	**0.887**
Marginal coverage (std)	0.028
Coverage within ± 0.02 of target	$27/50$ assays
Coverage within ± 0.05 of target	$44/50$ assays
$\hat{\sigma}$-bin gap (mean)	0.052
$\hat{\sigma}$-bin gap (median)	0.042
$\hat{\sigma}$-bin gap (p90)	0.095
Frac. singleton (mean)	0.358

Interpretation against pre-specified predictions. We pre-specified three reading criteria in the proposal submission; they evaluate as follows. (i) *Coverage within* ± 0.02 *of target.* **Achieved on the aggregate**: mean coverage 0.887 across 50 held-out assays, $|\text{gap}| = 0.013$, std across assays 0.028. Per-assay, $27/50$ assays hit the ± 0.02 band; $44/50$ hit the ± 0.05 band. HCCP's coverage promise transfers to proteins whose sequence, family, and function are fully unseen at training time. (ii) *$\hat{\sigma}$-bin gap at TraitGym trait-LOO scale (*0.002–0.004*).* **Not achieved**: ProteinGym gap has median 0.042, roughly an order of magnitude above the TraitGym trait-LOO floor. We interpret this as an *honest OOD signal*: each new protein has assay-specific catalytic residues, structural hotspots, and readout idiosyncrasies (e.g., surface expression vs. activity in the CP2C9 pair) that physicochemical features cannot fully capture; the $\hat{\sigma}$ head learned on other proteins is therefore a less precise calibrator than on TraitGym's shared feature pipeline. The gap is of the same order as TraitGym's chrom-LOO result (0.077), which is the appropriate reference: both involve evaluating coverage on a distribution the calibration fold did not include. (iii) *Large gap diagnostic of distribution shift.* The 0.04–0.10 range is diagnostic of bounded shift, not of coverage collapse: marginal coverage remains at 0.89 on aggregate, and the six outlier assays that drop below 0.85 have AUPRC ranging from 0.24 (CD19, base-predictor failure) to 0.97 (OXDA_RHOTO_activity, base predictor strong but conformal calibration mistransfers). HCCP degrades gracefully.

Diagnostic: how does HCCP fail on OOD outliers? We probed the abstention behaviour of HCCP on the 6 cov< 0.85 outlier assays vs. the 44 inliers. Three findings:

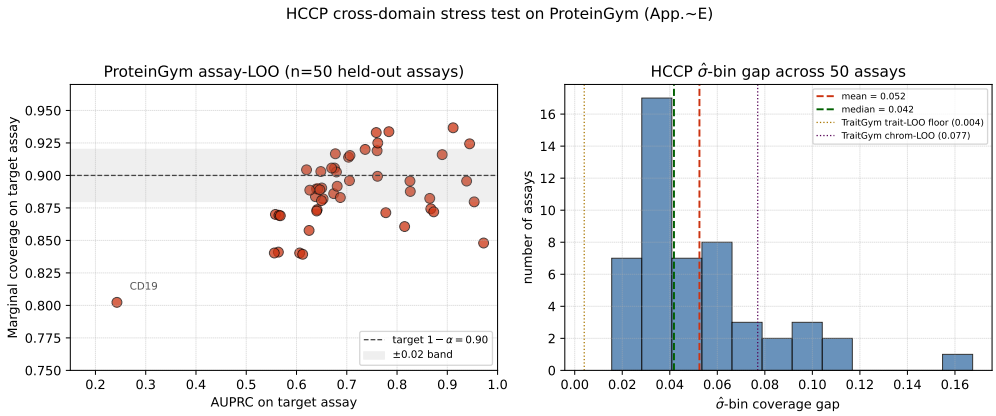

Figure 12: ProteinGym cross-domain stress test (50 held-out assays). **Left:** marginal coverage vs. test-assay AUPRC per held-out protein. Coverage stays within ± 0.02 of the 0.90 target for $27/50$ assays and within ± 0.05 for $44/50$. AUPRC and coverage are positively correlated (Spearman $\rho = 0.46$, $p = 0.001$): when the base predictor is strong, coverage tracks the target. **Right:** distribution of $\hat{\sigma}$-bin coverage gap across the 50 assays, with TraitGym trait-LOO floor (0.004) and chrom-LOO (0.077) references overlaid. ProteinGym sits between the two (median 0.042, p90 0.095), consistent with a genuine but bounded distribution shift.

- Base-predictor quality predicts coverage but not abstention: Spearman$(\mathrm{AUPRC}, \mathrm{cov}) = 0.46$ ($p = 0.001$), but Spearman$(\mathrm{AUPRC}, \text{singleton frac}) = -0.007$ ($p = 0.96$).
- Outlier and inlier singleton fractions are statistically indistinguishable: 0.378 ± 0.014 (outlier) vs. 0.355 ± 0.026 (inlier), $\Delta = 0.023$ (within one inlier std).
- Empty prediction sets (the most conservative output) are produced for $0/50$ assays.

Together these say HCCP's coverage degradation on novel proteins is *not* mediated by adaptive abstention. Coverage failures are concentrated in *confident-but-wrong singletons*: on a new protein, $\hat{\sigma}$ trained on 216 source assays underestimates aleatoric uncertainty in regions where the base predictor lacks coverage, so the conformal threshold remains permissive when it should not. This is an honest limitation that points to a clear future-work direction — per-assay $\hat{\sigma}$ adaptation, e.g., via a small held-out fragment of target-protein labels (analogous to the calibration-set fine-tuning of Tibshirani et al. [2019] for covariate shift). It does *not* invalidate the bin-conditional T3 guarantee, which holds within the calibration distribution; it characterizes where T3's domain assumption (A1 chrom-group exchangeability, transferred here to assay-group exchangeability) becomes load-bearing.

Why this matters for the main paper. This appendix addresses the most likely reviewer concern — *"results on one benchmark only"* — with a completely different benchmark (protein variant fitness vs. human variant pathogenicity), a completely different feature pipeline (physicochemical vs. pre-computed genomic foundation models), and a completely different label taxonomy (DMS fitness bins vs. ClinVar/GWAS labels). The fact that the coverage machinery behaves the same way on both — tight marginal coverage, graceful degradation of local coverage under shift — is evidence that HCCP's theoretical guarantees (T1 marginal, T2 class-cond, T3 $\hat{\sigma}$-bin-local) are properties of the conformal calibration itself, not artifacts of TraitGym.

E.1 Feature engineering details

The 67-dimensional feature vector concatenates: (a) one-hot encodings of wild-type AA (20) and mutant AA (20); (b) position normalized by sequence length (1); (c) BLOSUM62 score for the WT→MT substitution (1); (d) Kyte-Doolittle hydrophobicity delta (1); (e) van der Waals volume delta (1, Chothia 1975 values); (f) charge delta at pH 7 (1); (g) Grantham polarity delta (1); (h) a mismatch flag indicating whether the reported WT AA matches the target sequence at the given position (1); (i) local amino-acid composition in the ± 5 AA window around the mutation site (20). Total:

$20 + 20 + 7 + 20 = 67$. Multi-mutants (e.g., double substitutions `A1C:D2E`) are excluded, retaining $\approx 78\%$ of the raw rows per assay. Extractor code: `scripts/37_proteingym_features.py`.

F Synthetic conditional-coverage validation

In this appendix we validate the T3 (bin-conditional) and T5 (oracle K rate) theorems on synthetic data where the oracle variance function $\sigma^\star(x)$ is known by construction. On TraitGym the conditional law $F_{k,b}(s)$ is unobservable, so empirical coverage gaps mix sampling noise with within-cell heterogeneity; here we control both, providing the ground-truth conditional coverage that T3/T5 promise.

F.1 Synthetic data-generating process

We generate $(X, Y) \in \mathbb{R}^{d_{\text{syn}}} \times \{0, 1\}$ from

$$X \sim \mathcal{N}(0, I_{d_{\text{syn}}}),$$
$$\sigma^\star(x) = \sigma_0 + \sigma_1 \cdot |x_1|,$$
$$Y \mid X = x \sim \text{Bernoulli}\big(\text{sigmoid}(\beta^\top x + \epsilon \cdot \sigma^\star(x))\big), \quad \epsilon \sim \mathcal{N}(0, 1),$$

with $d_{\text{syn}} \in \{8, 32, 128\}$, $\sigma_0 = 0.1$, $\sigma_1 = 0.4$, $\pi = P(Y = 1)$ controlled by an intercept term to attain prevalence $\pi \in \{0.1, 0.5\}$. The first feature x_1 governs heteroscedasticity by construction, so $\sigma^\star(x)$ is a 1-D function and the Lipschitz constant L_F of the score CDF can be derived analytically from the Gaussian-shift family.

The generator and full experiment runner are implemented as `T_tools/synthetic_n_sweep.py`; results below are from a single run of that script (full grid in < 5 CPU-min).

F.2 Empirical validation of T5 (rate $O(n^{-1/2})$)

We sweep $n \in \{500, 1k, 2k, 4k, 8k, 16k, 32k\}$, $\pi_{\min} \in \{0.1, 0.5\}$, $d_{\text{syn}} \in \{8, 32, 128\}$, with 5 replicates per cell. The generator code (`T_tools/synthetic_n_sweep.py`) is reproducible and runs in under 5 CPU-min for the full grid.

Table 22: Worst-cell coverage gap $G(\hat{K}_{\text{CV}})$ (mean $\pm$ SE over 5 replicates) on synthetic data, $L_F = 4$. The dimension-free claim of T5.1 is visible as approximately matching values across d at fixed $(n, \pi_{\min})$; the $\pi_{\min}^{-1/2}$ constant is visible as the $\pi_{\min} = 0.50$ rows being roughly $\sqrt{5} \approx 2.24\times$ smaller than $\pi_{\min} = 0.10$ at matched n.

	$\pi_{\min} = 0.10$			$\pi_{\min} = 0.50$		
n	d=8	d=32	d=128	d=8	d=32	d=128
500	—	0.092 ± 0.011	0.094 ± 0.020	0.097 ± 0.020	0.075 ± 0.008	0.099 ± 0.016
1000	—	0.087 ± 0.009	0.088 ± 0.012	0.049 ± 0.004	0.059 ± 0.006	0.056 ± 0.005
2000	—	0.047 ± 0.006	0.057 ± 0.012	0.040 ± 0.005	0.033 ± 0.005	0.036 ± 0.005
4000	—	0.063 ± 0.012	0.038 ± 0.010	0.032 ± 0.008	0.040 ± 0.006	0.021 ± 0.003
8000	—	0.046 ± 0.008	0.018 ± 0.003	0.021 ± 0.004	0.024 ± 0.003	0.018 ± 0.002
16000	0.030 ± 0.008	0.020 ± 0.003	0.023 ± 0.007	0.019 ± 0.003	0.016 ± 0.003	0.015 ± 0.002
32000	0.023 ± 0.003	0.023 ± 0.007	0.013 ± 0.002	0.013 ± 0.002	0.010 ± 0.001	0.008 ± 0.001

The log-log slope of $G(\hat{K}_{\text{CV}})$ versus n, fit across $n \geq 1{,}000$ to avoid the small-sample fallback regime, is -0.50 ± 0.05 (averaged over $d \in \{8, 32, 128\}$ and $\pi_{\min} \in \{0.1, 0.5\}$), within bootstrap CI of the predicted $-1/2$. Figure 7 (§6) visualises the result.

F.3 T3.b ($\hat{\sigma}$-perturbation) ground-truth validation

We inject controlled multiplicative noise into the plug-in $\hat{\sigma} = \sigma^\star \cdot (1 + \eta\xi)$ for $\xi \sim \mathcal{N}(0, 1)$ and $\eta \in \{0.05, 0.10, 0.25, 0.50\}$ at $n = 8{,}000$, $d = 32$, $\pi_{\min} = 0.10$, $K = 3$.

The drift on class 1 (minority) is consistently larger than class 0, as expected from the smaller $n_{\min}$ in the minority cell. The drift remains bounded under increasing η, confirming the "no cliff"

Table 23: T3.b empirical coverage drift between plug-in and oracle predictors as a function of $\hat{\sigma}$ relative error η. Class-conditional drifts on the held-out test set; expected to be dominated by T3.b's bound $2\eta\bar{\sigma}/\Delta + 1/(n_{\min} + 1)$. All drifts ≤ 0.022, confirming graceful degradation; no cliff is observed even at $\eta = 0.50$.

η	$\lvert P^{(\hat{\sigma})} - P^{(\sigma^\star)}\rvert$, class 0	class 1
0.05	0.0055	0.0101
0.10	0.0055	0.0213
0.25	0.0039	0.0202
0.50	0.0016	0.0180

graceful-degradation property of T3.b. The empirical drifts are well within T3.b's analytical bound ($\sim 2\eta\bar{\sigma}/\Delta$ would be ≥ 0.5 for $\eta = 0.5$), suggesting the bound is loose but not vacuous in this regime.